

Lecture 02:

User Interaction Models

COMS 4170 • Spring 2026

Prof. Celeste Layne
Tuesday January 27, 2026



Interfaces display information in a way that helps users accomplish a goal.

```
i:Exit --PrevPg <Space>NextPg v:View Attachm. d:Del r:Reply j:Next ?:Help
624 Aug 03 T Martinez ( 37) Loans with tiny points are here now
625 0 Jul 01 R. Jackson ( 123) Loans with tiny rates are here now
626 Aug 05 Benjamin E. Mag ( 50) Long time no hear
627 May 17 Krista Aaron ( 44) long time no see....
628 0 Jun 03 Josiah House ( 35) Looking for a hot date tonight, tomorrow, or next week?
629 Jul 03 Brigitte I. Hay ( 63) Looking for a N.ew H.ome?
630 May 17 Joe Burns ( 58) Looking for you
631 Jun 01 Save in a poor ( 145) Low Rate Consolidation Mortgage Loan
632 + Jul 02 Igiel@virtualig ( 2) LowCost SoftWare OnCD
--Mutt: Mail/junk/spam [Msgs:950 Old:142 10M]--(subject/date)--(66%)
Date: Mon, 17 May 2004 03:40:09 +0100
From: Krista Aaron <Christinefeminine@highstream.com>
Subject: long time no see....

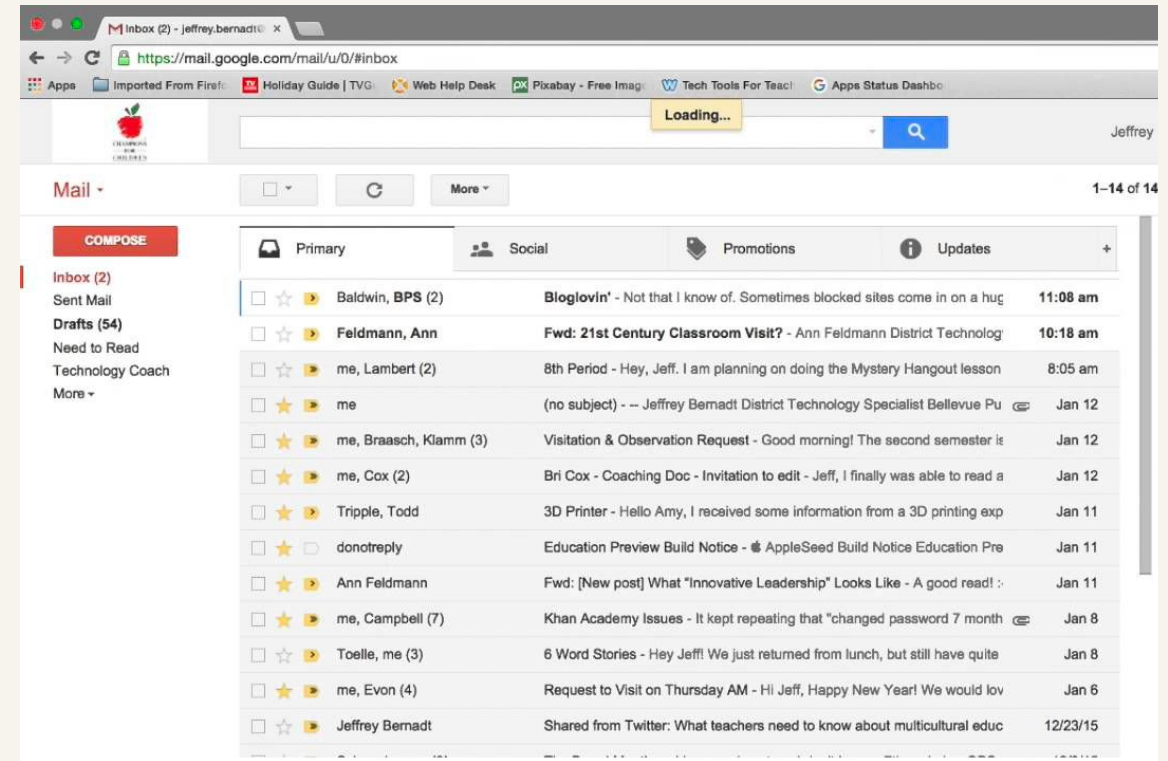
[-- Autoview using /usr/bin/links -force-html -dump '/tmp/mutt.html' --]
My name is Jen and I'm new to this dating thing. I've checked out your profile
you put up and it's interesting. :) I just want to get to know you a little
better if you don't mind, come check my profile out at:

    www.livejen.com/chat.html

I also got a webcam so we can make it interesting, anyways hope you get back to
me.
    bye :)

    gxsnkxxgnduvyjjwceudc.jobxs
    zcozccrociesbehgbpow
    rnxlfu.jnqpbliipdkgwuyqofracsz
    xmqawbxsbjrppoibvlpfhqouldtp
    bixhghvrxxtqgfeocofzyzb
    hugzffaffulsklpzhrfxbtt
    btpztlfotqmmoaiwlosqv
-- 627/950: Krista Aaron long time no see.... --(69%)
Key is not bound. Press '?' for help.
```

Pine text-based email client, 1989

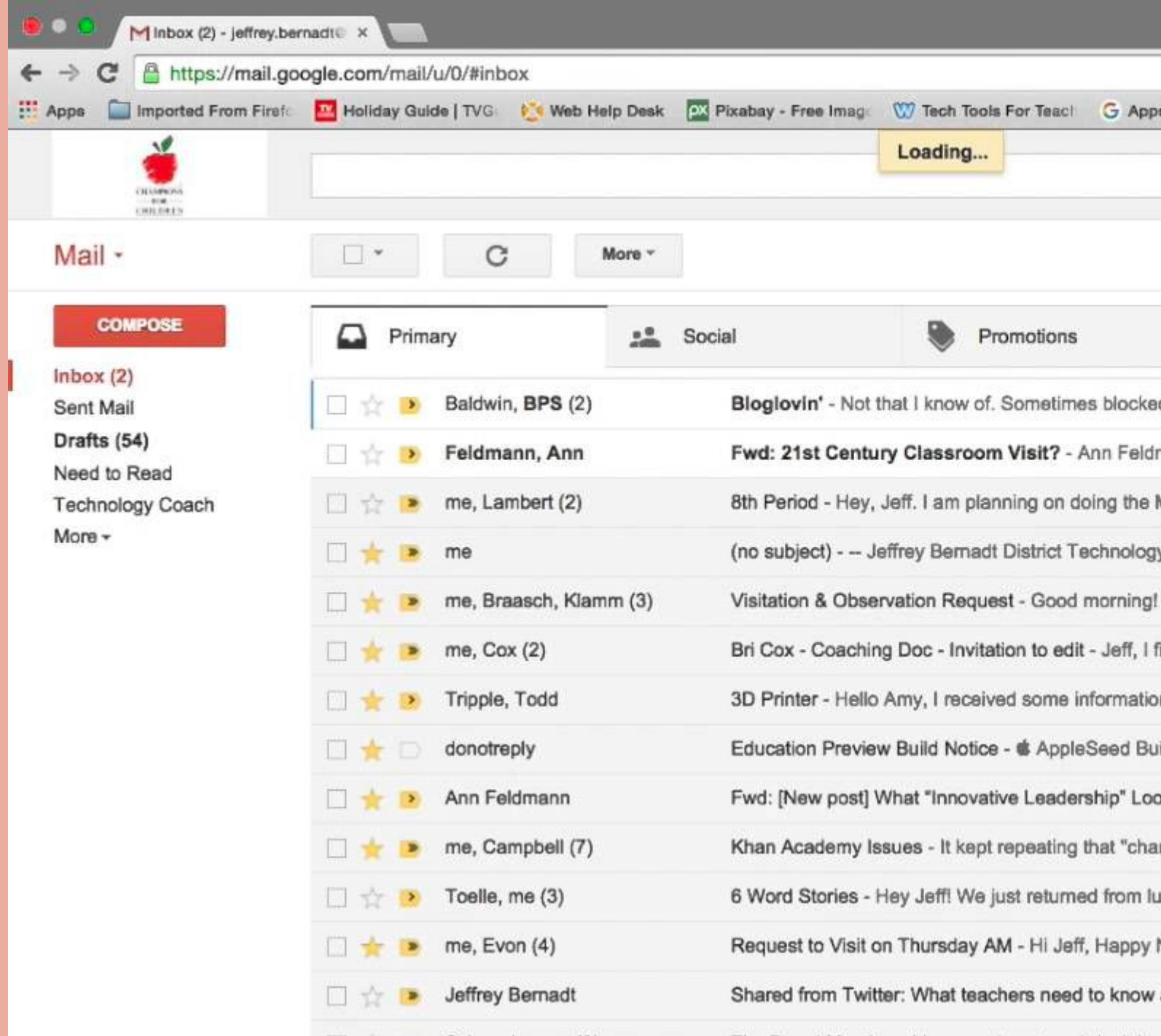


Gmail browser-based email service, 2004



What is the primary goal of this user interface?

To read and respond to all emails received.



What is the primary goal of this user interface?

To purchase books.

The screenshot displays the Amazon Books homepage. At the top, the Amazon Prime logo is visible on the left, and a search bar with a magnifying glass icon is on the right. Below the logo, a navigation bar includes links for Departments, Browsing History, Lydia's Amazon.com, Today's Deals, Gift Cards, Registry, Sell, and Help. A secondary navigation bar lists categories like Books, Advanced Search, New Releases, Amazon Charts, Best Sellers & More, The New York Times® Best Sellers, Children's Books, Textbooks, Textbook Rentals, and Sell Us Your Books.

The main content area is divided into several sections:

- Popular in Books:** A list of categories including Award Winners, Bargain Books, Best Books of the Month, Best Books of 2017, Books in Spanish, Children's Books, Deals in Books, and Top 20 Lists in Books.
- More in Books:** A list of categories including 100 Books to Read in a Lifetime, Amazon Book Review Blog, Amazon Books on Facebook, Amazon Books on Twitter, and Amazon First Reads.
- Amazon Kindle:** A list of categories including Kindle eBooks, Kindle Deals, Kindle Unlimited, Prime Reading, and Spanish Kindle eBooks.
- Textbooks:** A list of categories including Textbooks Store, Textbook Rentals, Sell Us Your Books, and Kindle eTextbooks.

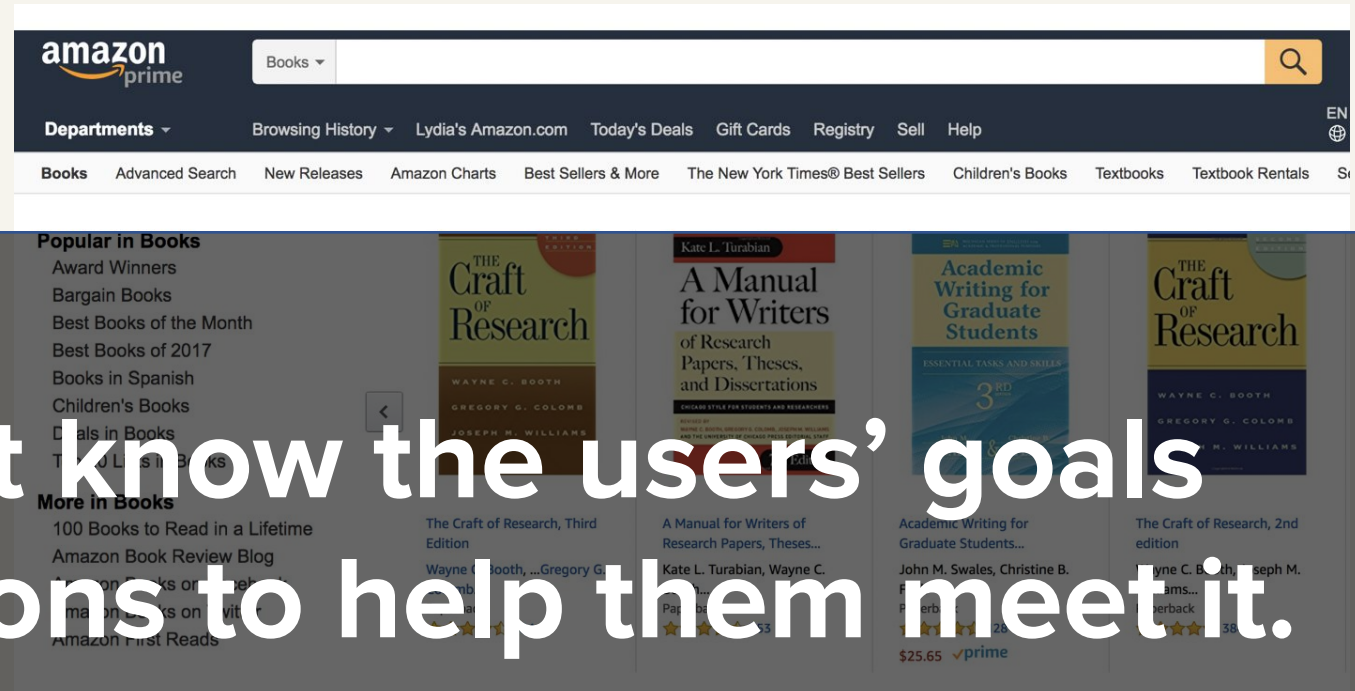
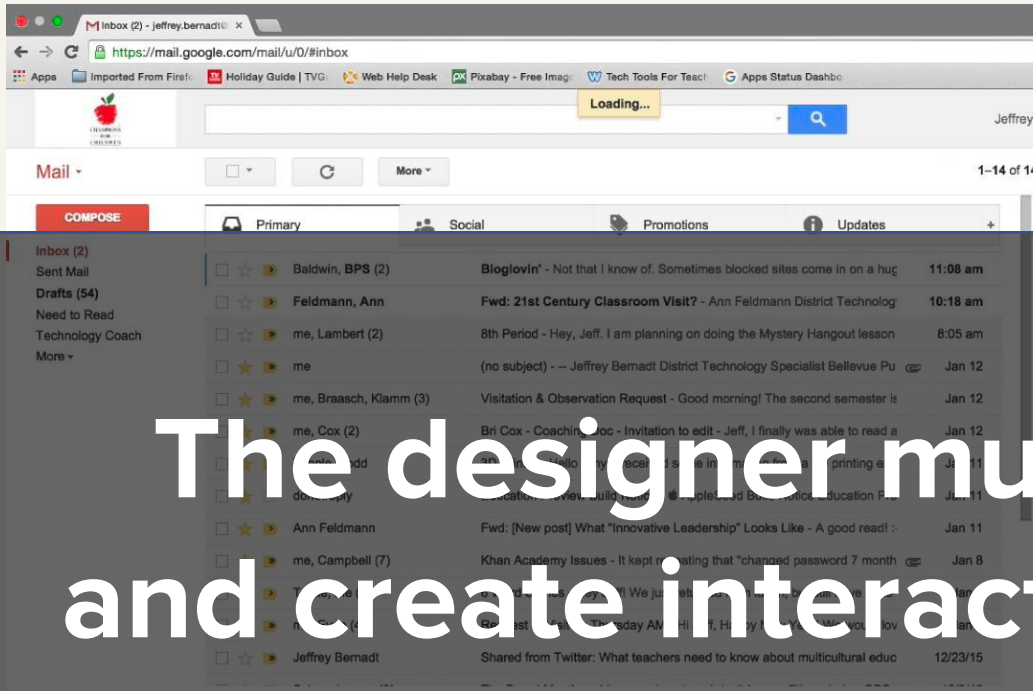
The central part of the page features a carousel of book covers with a left arrow button. The books shown are:

- The Craft of Research, Third Edition** by Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams. It has a 4.5-star rating and 384 reviews.
- A Manual for Writers of Research Papers, Theses, and Dissertations** by Kate L. Turabian and Wayne C. Booth. It is the 7th Edition, has a 4.5-star rating and 753 reviews, and is available as a paperback.
- Academic Writing for Graduate Students** by John M. Swales and Christine B. Feak. It is the 3rd Edition, has a 4.5-star rating and 128 reviews, and is available as a paperback for \$25.65 with Prime.

Below the carousel, there is a section titled **More from the Amazon Book Editors** with three featured book lists:

- Best Books of the Month Top 10:** Literature & Fiction, Kids & Teens Books, Mystery & Thriller. A link to "See all categories" is provided.
- Celebrity Picks 2017:** Sheryl Sandberg, John Green, Gabrielle Union, Denis Leary. A link to "See all Celebrity Picks" is provided.
- Best Books of 2017 Top 100 Print Books:** Fiction, Children's Books, Biographies & Memoirs. A link to "See all" is provided.

Users interact with a system to accomplish a goal



The designer must know the users' goals and create interactions to help them meet it.

To read and respond to all email.

To buy a book

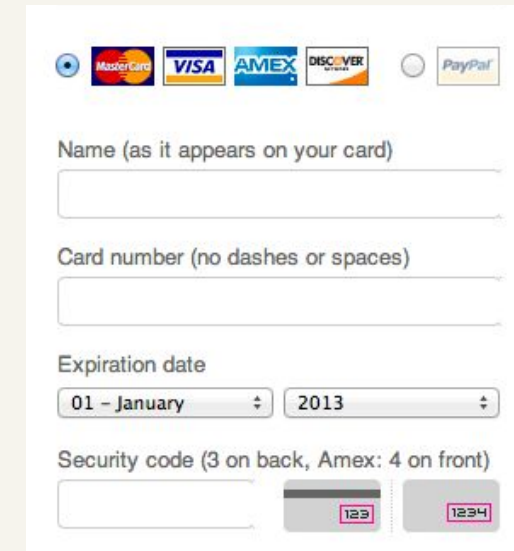
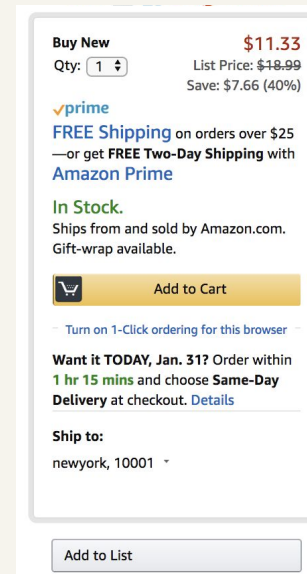


Large goals must be broken up into smaller goals

Goal: Purchase a book

Subgoals:

1. Find the book
2. Add to cart
3. Enter payment info
4. Place the order



Goal: Purchase a book

Subgoal: Search for a book

Interaction: Type its name, click on the image of it

amazon [Try Prime](#)

All ▾ design of everyday things 🔍

Shop Valentine's Day Deals

Departments ▾ Your Amazon.com Today's Deals Gift Cards Registry Sell Help

EN ▾ Hello, Sign in **Account & Lists** ▾ Orders Try Prime ▾  0 Cart

1-16 of over 6,000 results for "design of everyday things" Sort by Relevance ▾

☐ **FREE Shipping**

All customers get FREE Shipping on orders over \$25 shipped by Amazon

Ad feedback 

Show results for

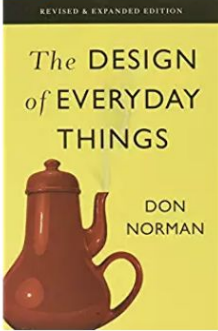
Books

- Popular Applied Psychology
- Medical Applied Psychology
- Business & Money
- Industrial & Product Design
- Retailing Industry
- [See more](#)

Kindle Store

- Applied Psychology
- Retailing Industry
- Consumer Behavior
- Business & Investing
- Web Site Design
- [See more](#)
- [See All 15 Departments](#)

Best Seller



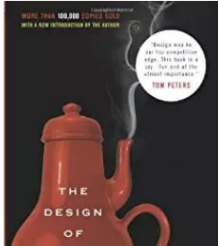
The Design of Everyday Things: Revised and Expanded Edition Nov 5, 2013
by Don Norman

Paperback
\$11.33 ~~\$18.99~~  **prime**
Get it by **TODAY, Jan 31**
FREE Shipping on eligible orders

More Buying Choices
\$4.93 (129 used & new offers)

Kindle Edition
\$11.99
Get it **TODAY, Jan 31**

★★★★★ ▾ 277



The Design of Everyday Things Sep 19, 2002
by Donald A. Norman

Paperback
\$3.22 (147 used & new offers)

Audible Audio Edition
\$0.00
Free with Audible trial

★★★★★ ▾ 301



Goal: Purchase a book

Subgoal: Add to cart

Interaction: Click “Add to cart” button

amazon

Try Prime

All

Shop Valentine's Day Deals

Departments

Your Amazon.com

Today's Deals

Gift Cards

Registry

Sell

Help

EN

Hello. Sign in

Account & Lists

Orders

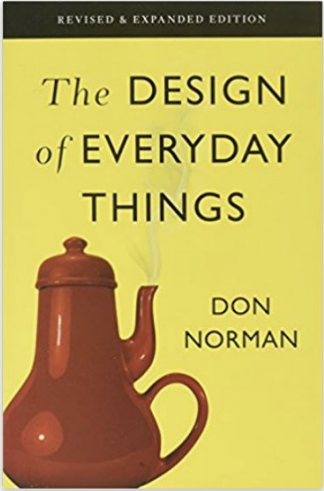
Try Prime

1 Cart

Back to search results for "design of everyday things"

The Design of Everyday Things: Revised and Expanded Edition and over one million other books are available for Amazon Kindle. [Learn more](#)

Look inside



REVISED & EXPANDED EDITION

The DESIGN of EVERYDAY THINGS

DON NORMAN

See all 2 formats and editions

Kindle

\$11.99

Read with Our Free App

Paperback

\$11.33

62 Used from \$4.93

62 New from \$5.33

5 Collectible from \$9.00

Even the smartest among us can feel inept as we fail to figure out which light switch or oven burner to turn on, or whether to push, pull, or slide a door. The fault, argues this ingenious—even liberating—book, lies not in ourselves, but in product design that ignores the needs of users and the principles of cognitive psychology. The problems range from ambiguous and hidden controls to arbitrary relationships between controls and functions, coupled with a lack of feedback or other assistance and unreasonable demands on memorization. *The Design of Everyday Things* shows that good, usable design is possible. The rules

Read more

The Amazon Book Review

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Share

<Embed>

Buy New

\$11.33

Qty: 1

List Price: ~~\$18.99~~

Save: \$7.66 (40%)

prime

FREE Shipping on orders over \$25

—or get FREE Two-Day Shipping with Amazon Prime

In Stock.

Ships from and sold by Amazon.com.

Gift-wrap available.

Add to Cart

Turn on 1-Click ordering for this browser

Want it TODAY, Jan. 31? Order within 1 hr 15 mins and choose Same-Day Delivery at checkout. [Details](#)

Ship to:

newyork, 10001



Goal: Purchase a book

Subgoal: If they are done with the cart, checkout

Interaction: Click “Proceed to checkout” button

amazon

Try Prime

All

Q

Shop Valentine's Day Deals

Departments

Your Amazon.com

Today's Deals

Gift Cards

Registry

Sell

Help

EN

Hello. Sign in

Account & Lists

Orders

Try Prime

Cart

Books

Advanced Search

New Releases

Amazon Charts

Best Sellers & More

The New York Times® Best Sellers

Children's Books

Textbooks

Textbook Rentals

Sell Us Your Books

Best Books of the Month

✓



Added to Cart

Cart subtotal (1 item): \$11.33

Add \$13.67 of eligible items for [FREE Shipping](#) or sign up for [Amazon Prime](#) and get faster, FREE Two-Day Shipping.

Cart

Proceed to checkout (1 item)



Get a **\$50 Amazon.com Gift Card instantly** upon approval for the **Amazon Rewards Visa Card**

Current

\$ 11.33

Total:

Savings: - \$ 50.00

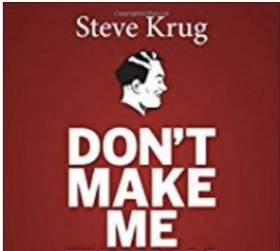
Cost After \$ 0.00

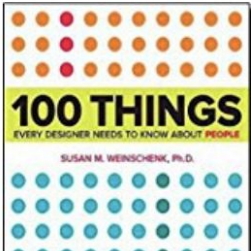
Savings: \$ 38.67

Remaining:

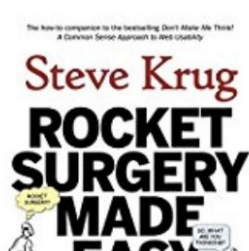
Apply now

Customers also bought these highly rated items











Goal: Purchase a book

Subgoal: Enter payment information

Interaction: Click, type, move cursor, click “Continue”





[SIGN IN](#) [SHIPPING & PAYMENT](#) [GIFT OPTIONS](#) [PLACE ORDER](#)



Secure Payment Info



Name (as it appears on your card)

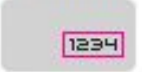
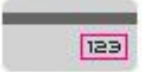
Card number (no dashes or spaces)

Expiration date

01 – January

2013

Security code (3 on back, Amex: 4 on front)



Continue

You can review this order before it's final.



Goal: Purchase a book

Subgoal: If the information is correct, place order

Interaction: Click “Place your order”



[SIGN IN](#) [SHIPPING & PAYMENT](#) [GIFT-WRAP](#) [PLACE ORDER](#)

Review Your Order

By placing your order, you agree to Amazon.com's [privacy notice](#) and [conditions of use](#)

Shipping Address:
Chris Customer
742 EVERGREEN TERRACE
SPRINGFIELD, WV 20025
United States
Phone: 1234567890 [Change](#)

Billing Information:
Rewards Points
 ending in 1234 [Change](#)
Billing Address:
Same as shipping address [Change](#)

Gift Cards & Promotional Codes:
 [Apply](#)

Estimated delivery: Sept. 26, 2011
**Apple iPad Camera Connection Kit (MC531ZM/A)**
\$29.95
 **Quantity: 1** [Change](#)
Sold by: -Media-Mart
[Add gift options](#)

Choose your  shipping speed:
☐ FREE Standard Shipping (3-5 business days)
☒ FREE Two-Day Shipping **—get it Monday, September 26**
☐ \$3.99/item One-Day Shipping **—get it Saturday, September 24**

[Place your order](#)

Order Summary

Items:	\$29.95
Shipping & Handling:	\$0.00
Total Before Tax:	\$29.95
Estimated Tax To Be Collected:	\$0.00
Rewards Points	-\$4.58

Order Total: \$25.37

How are shipping costs calculated?
Amazon Prime Shipping has been applied to the eligible items in your order.



The designer must know the users' goals create the subgoals and interactions to meet it.

Goal: Purchase a book

Subgoal:

Find it

Add to cart

Enter payment info

Place order

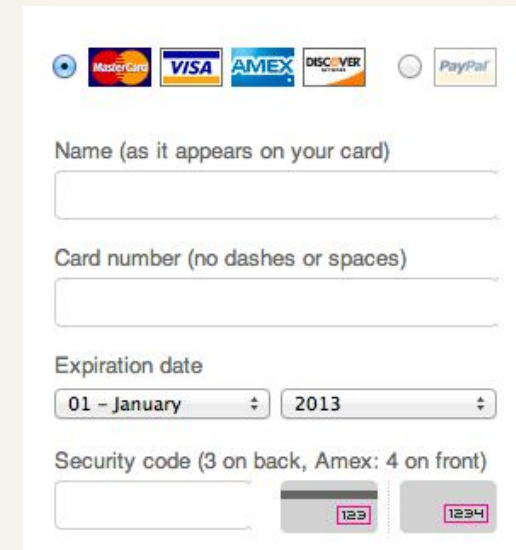
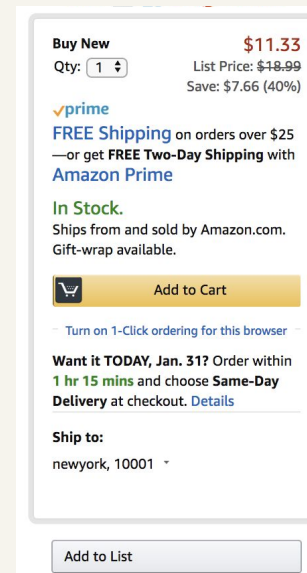
Interaction:

Type and click

Click

Type, click, point

Click



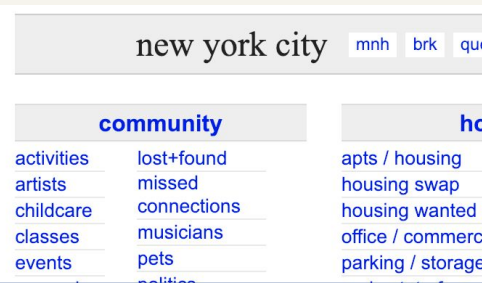
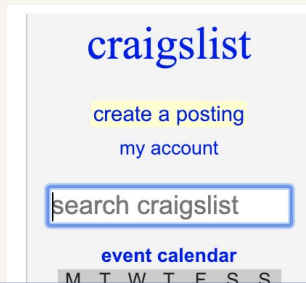
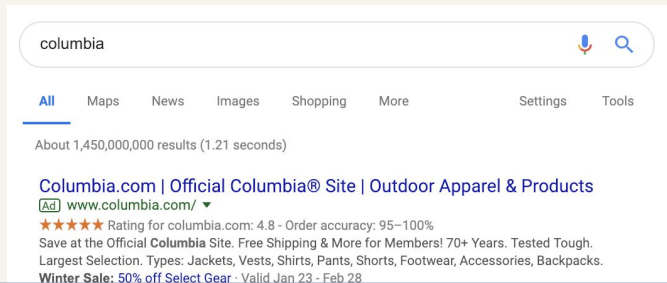
The designer must know the users' goals
create the subgoals and interactions to meet it.

**but sometimes on
Amazon, I don't
have a goal.**

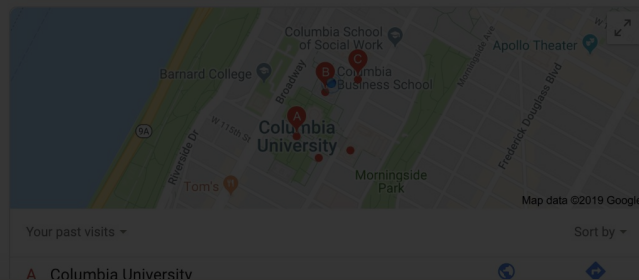
**I'm just
browsing** 🤔



Are these sites “just for browsing”?



No. Although people do browse,
Browsing is secondary.
their primary goal is to find or discover
something.



The designer must know the users goals and create interactions that meet those goals.

amazon Try Prime

All ▾ design of everyday things 🔍

Shop Valentine's Day Deals

Departments ▾ Your Amazon.com Today's Deals Gift Cards Registry Sell Help

EN ▾ Hello. Sign in Account & Lists ▾ Orders Try Prime ▾  Cart

1-16 of over 6,000 results for "design of everyday things" Sort by Relevance ▾

☐ FREE Shipping

All customers get FREE Shipping on orders over \$25 shipped by Amazon

Show results for

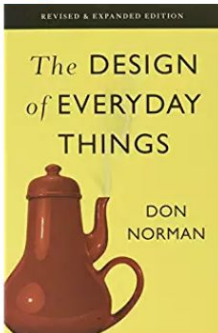
Books

- Popular Applied Psychology
- Medical Applied Psychology
- Business & Money
- Industrial & Product Design
- Retailing Industry
- See more

Kindle Store

- Applied Psychology
- Retailing Industry
- Consumer Behavior
- Business & Investing
- Web Site Design
- See more
- See All 15 Departments

Best Seller



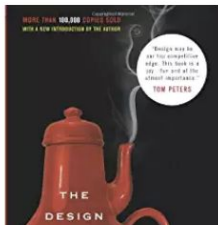
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Ad feedback 



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Low-level Interactions

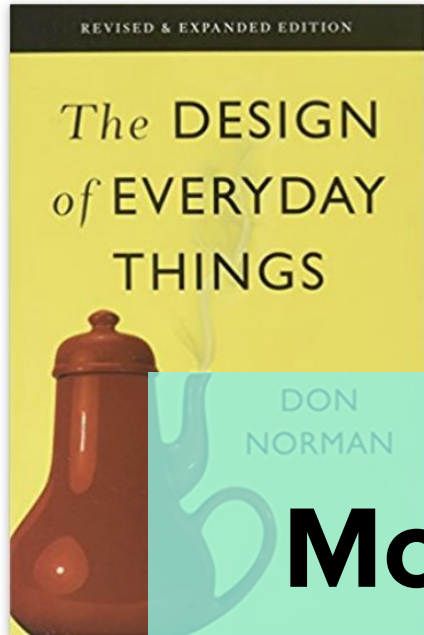


What are interactions on this page?

◀ Back to search results for "design of everyday things"

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The Design of Everyday Things: Revised and Expanded Edition Paperback –

November 5, 2013

by [Don Norman](#) ▾ (Author)

★★★★★ ▾ 277 customer reviews

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62 New from \$5.33

5 Collectible from \$9.00

Move

Click

Even the smartest among us can feel inept as we fail to figure out which light switch or oven burner to turn on, or whether to push, pull, or slide a door. The seemingly obvious, yet ingenious—even liberating—book, lies not in ourselves, but in product design that ignores the needs of users and the principles of cognitive psychology. The problems range from ambiguous and hidden controls to arbitrary relationships between controls and functions, coupled with a lack of feedback or other assistance and unreasonable demands on memorization. *The Design of Everyday Things* shows that good, usable design is possible. The rules

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\$11.33

Qty:

List Price: ~~\$18.99~~

Save: \$7.66 (40%)

✓prime

FREE Shipping on orders over \$25
—or get **FREE Two-Day Shipping** with
Amazon Prime

In Stock.

Ships from and sold by Amazon.com.

Gift-wrap available.



Add to Cart

Type

[Ordering for this browser](#)
Can't wait TODAY, Jan. 31? Order within
1 hr 15 mins and choose **Same-Day**
Delivery at checkout. [Details](#)

Ship to:

newyork, 10001 ▾

[Add to List](#)

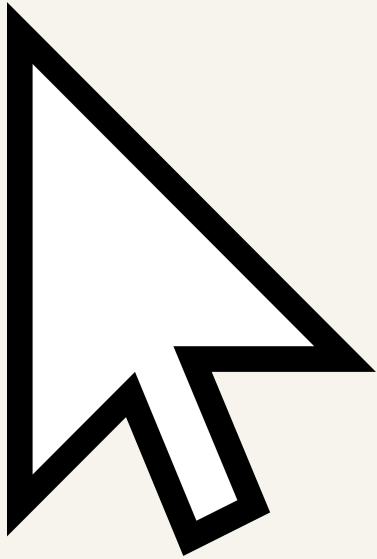
[See all 4 images](#)

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Interaction: Moving & Clicking



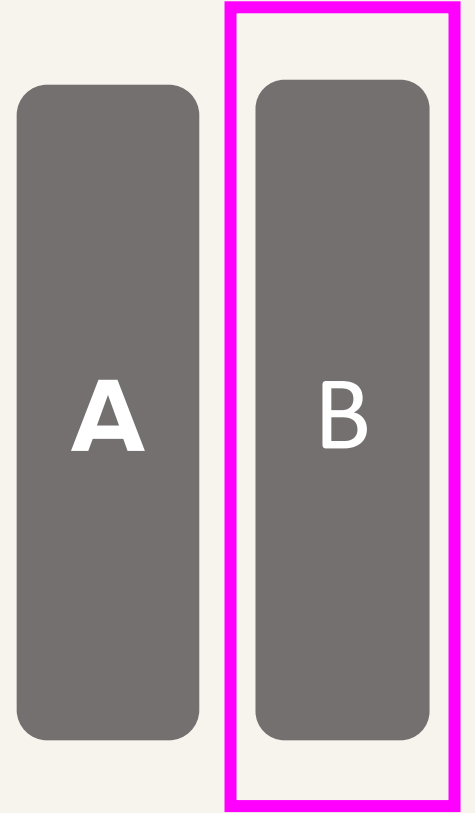
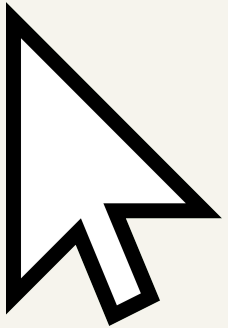
Which button is faster to click?

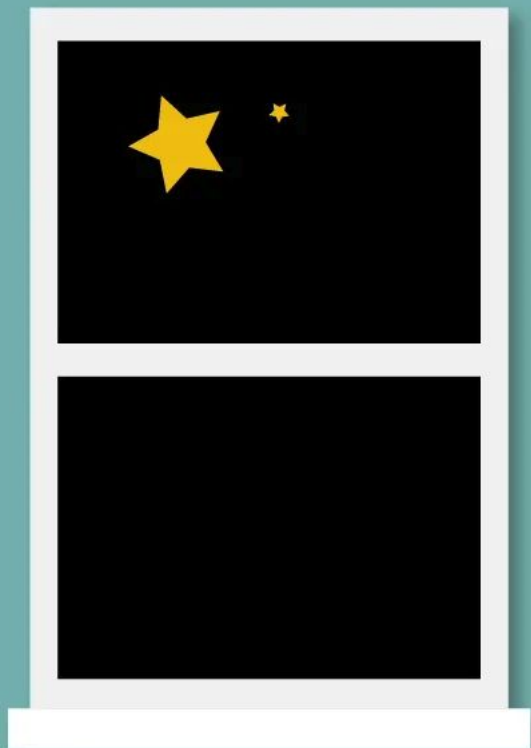


Which button is faster to click?

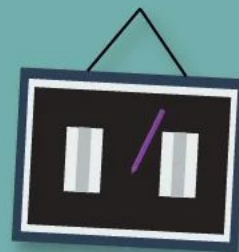


Which button is faster to click?

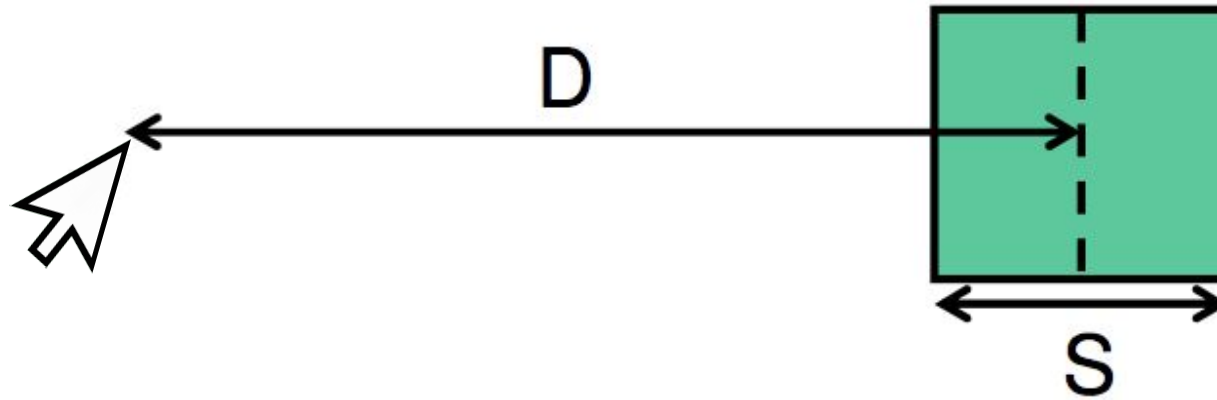




Fitts's Law

$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$


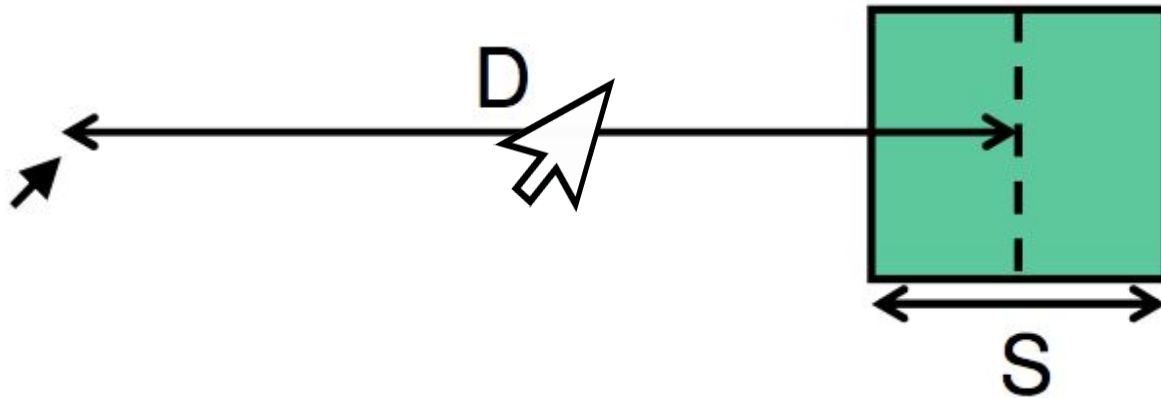
Fitts's Law



Time to move your
pointer to a target

$$= a + b * \log \left(\frac{2D}{S} \right)$$

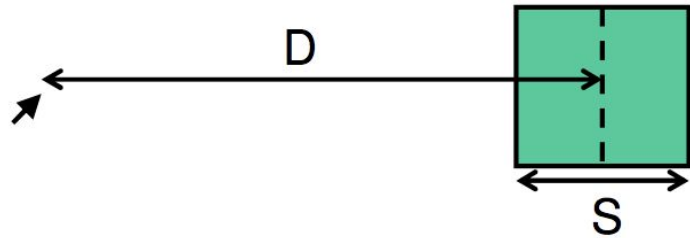
How does Fitts's Law work?



$$T = a + b * \log \left(\frac{2D}{S} \right)$$

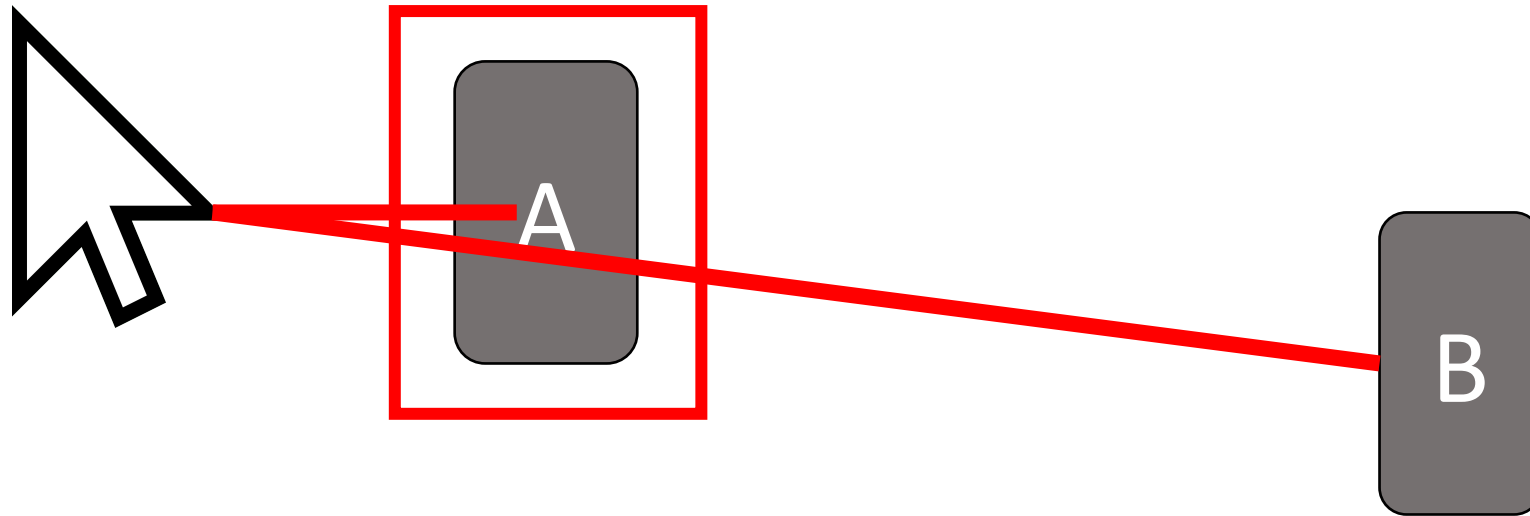
1. **T**, amount of time required to complete the movement
2. **D is the distance** from your cursor to the target
3. **S is the size** (width) of the target
4. a and b are constants

Using Fitts' law, why is A faster to click?

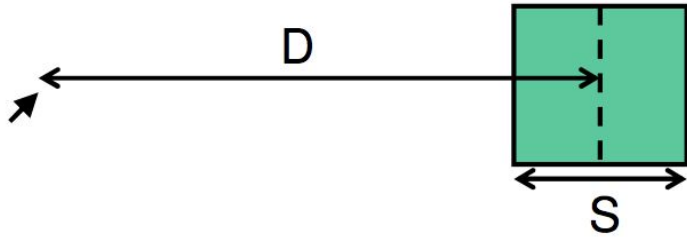


$$= a + b * \log \left(\frac{2D}{S} \right)$$

D is small;
therefore, the time is
lower.

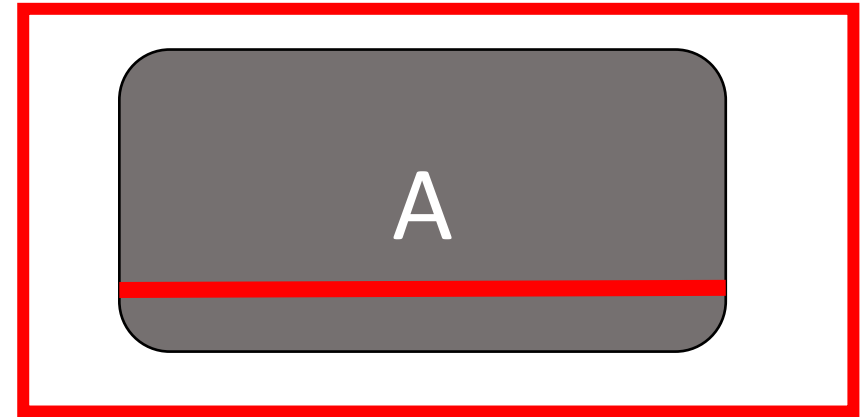
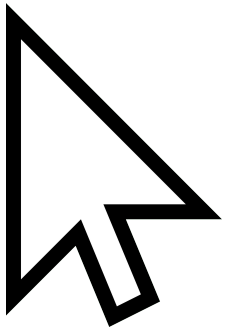


Using Fitts' law, why is A faster to click?

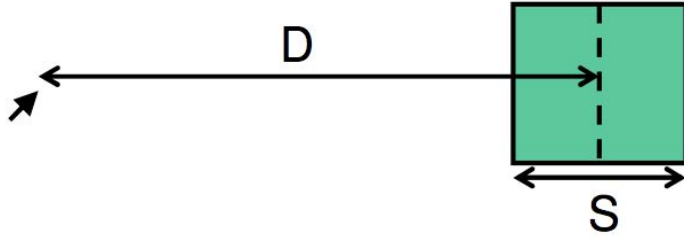


$$= a + b * \log \left(\frac{2D}{S} \right)$$

S is bigger;
therefore the time is
lower.

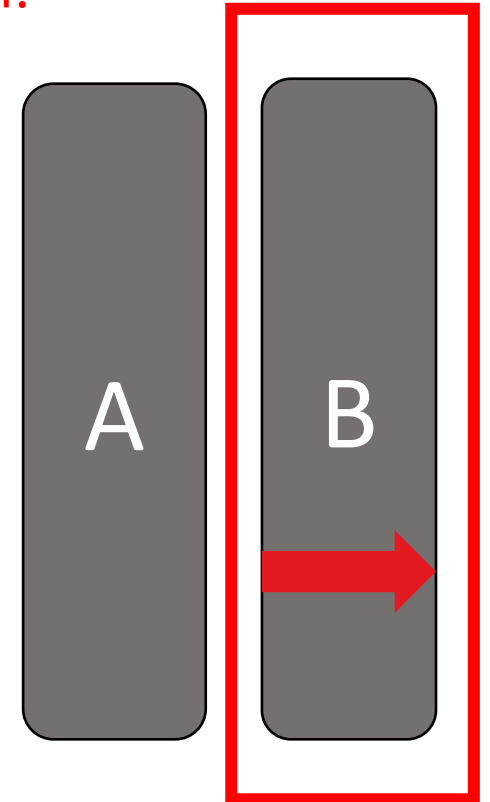
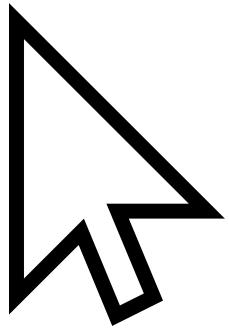


Using Fitts' law, why is B faster to click?

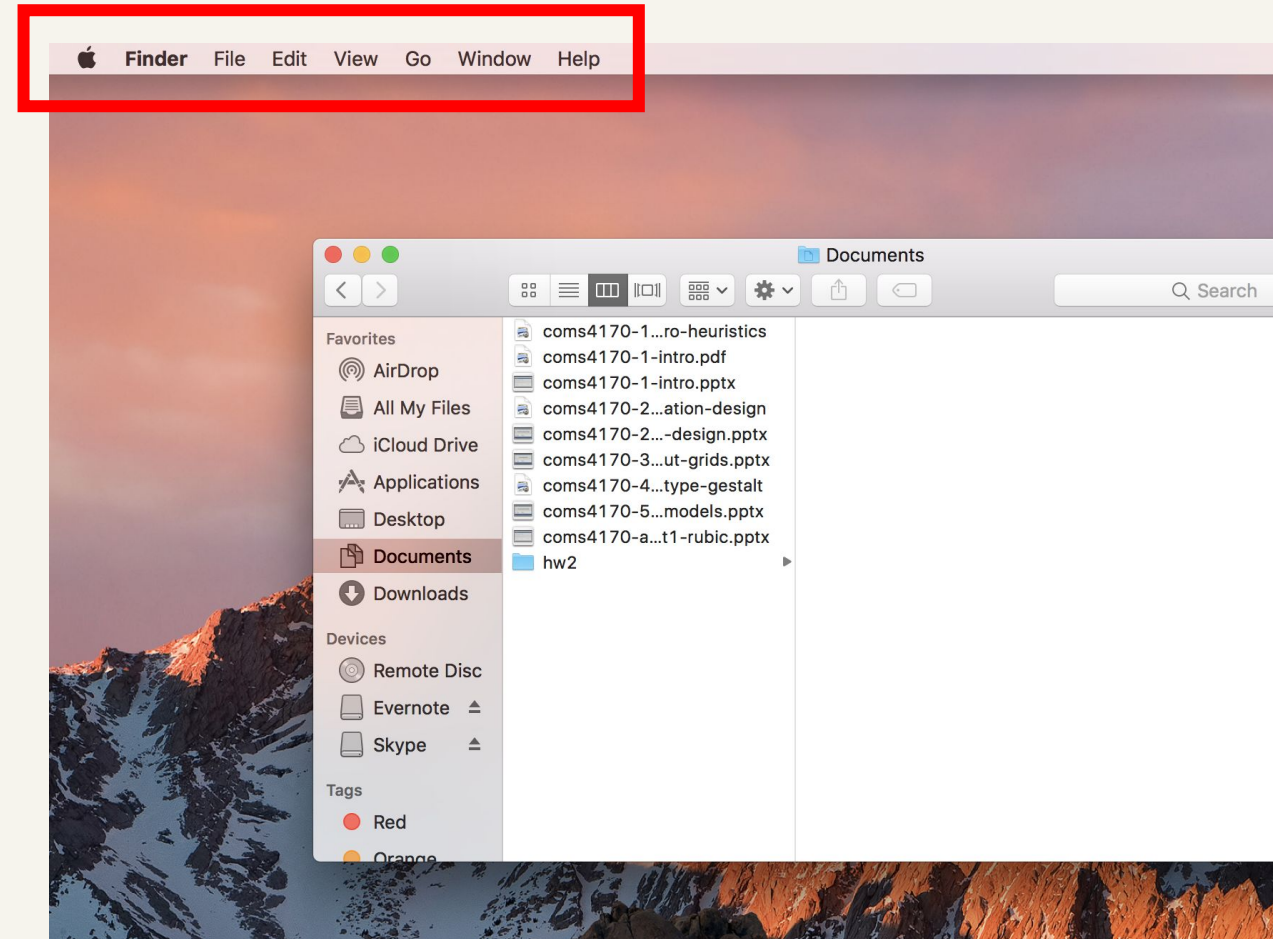
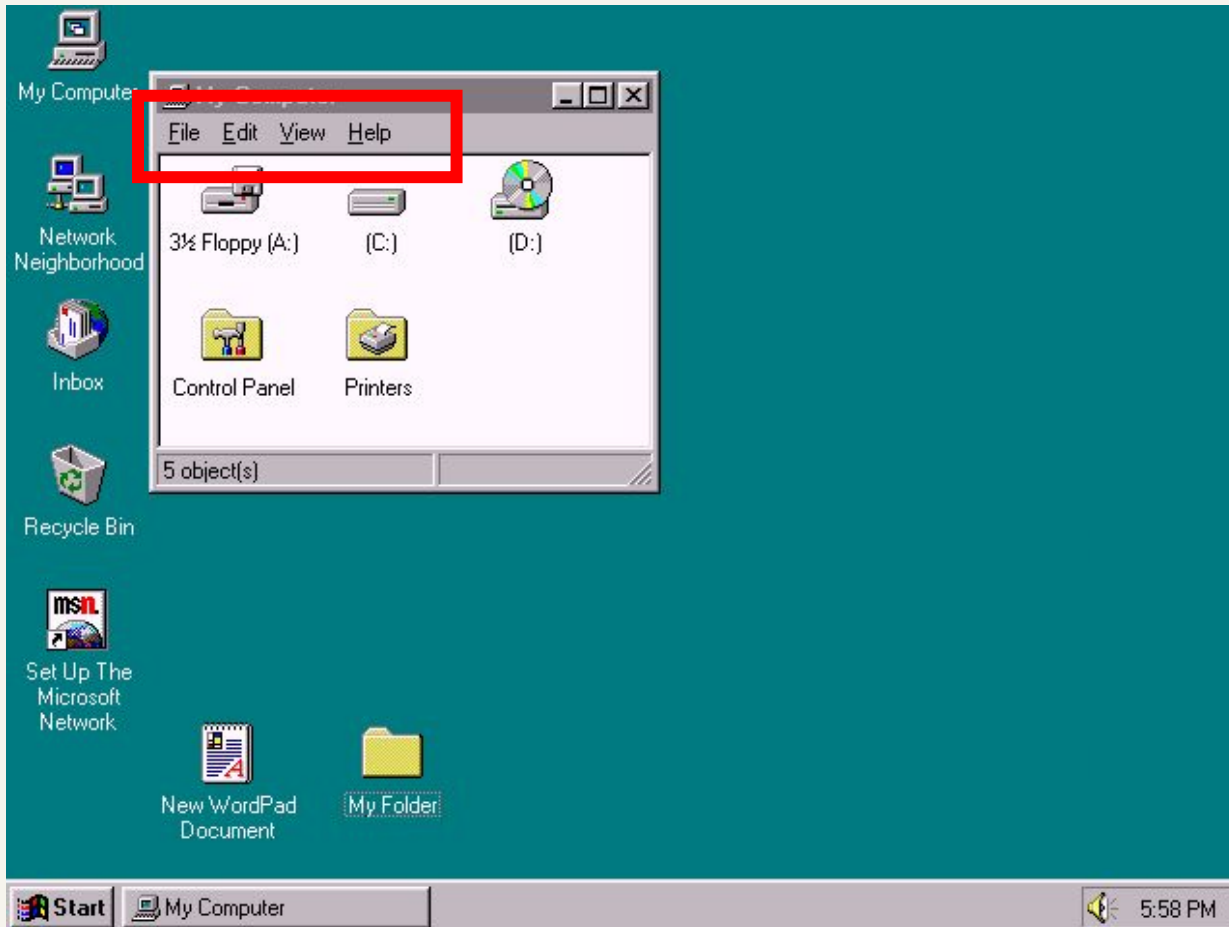


$$= a + b * \log \left(\frac{2D}{S} \right)$$

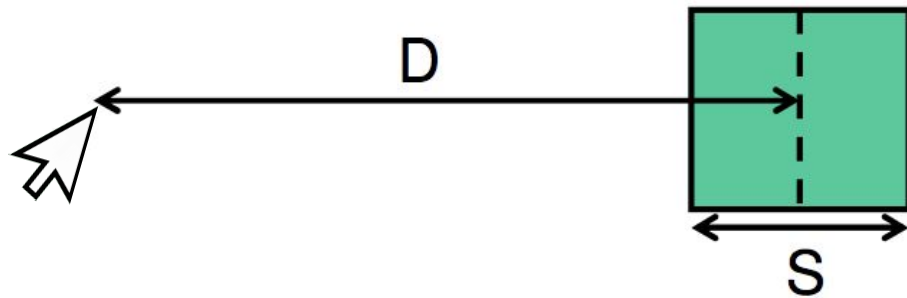
S is bigger (infinite);
therefore, the time is
lower.



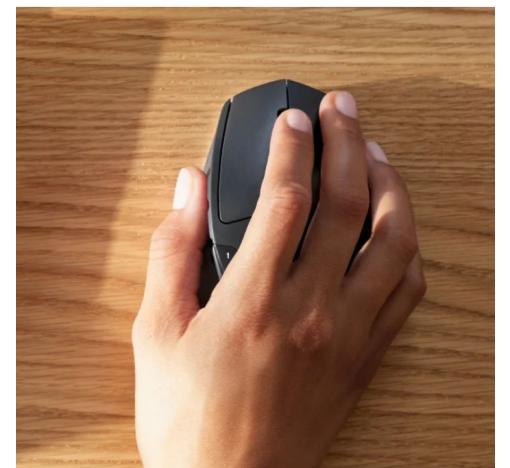
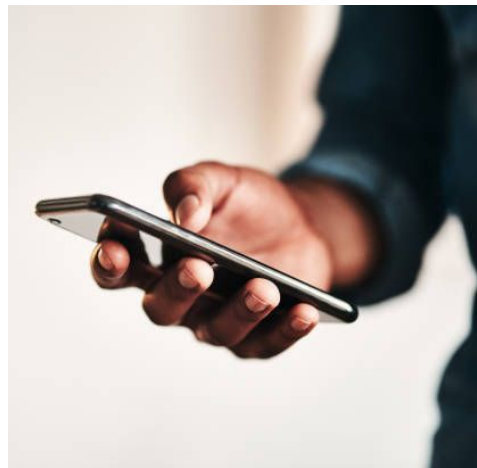
This is why iOS designed the menu to touch the edge of the screen



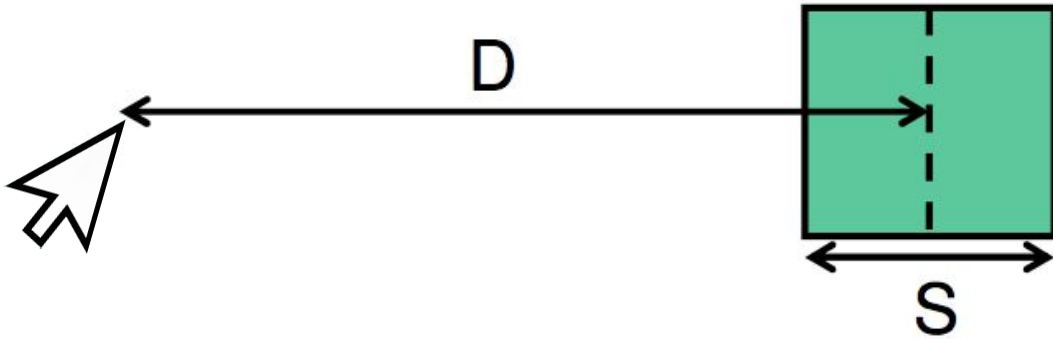
Fitts's Law: what are a and b?



$$\text{time} = a + b * \log \left(\frac{2D}{S} \right)$$

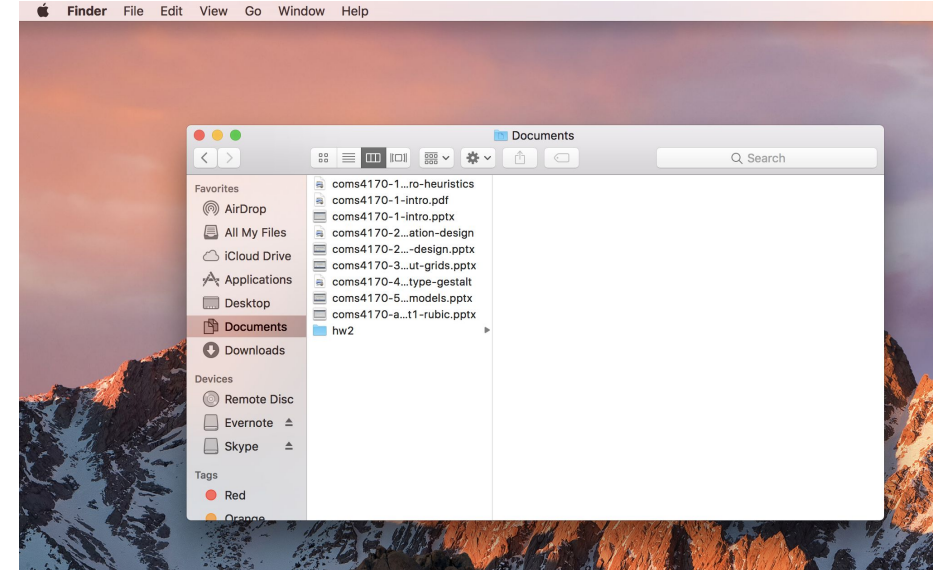


Fitts's Law: Time to move the pointer



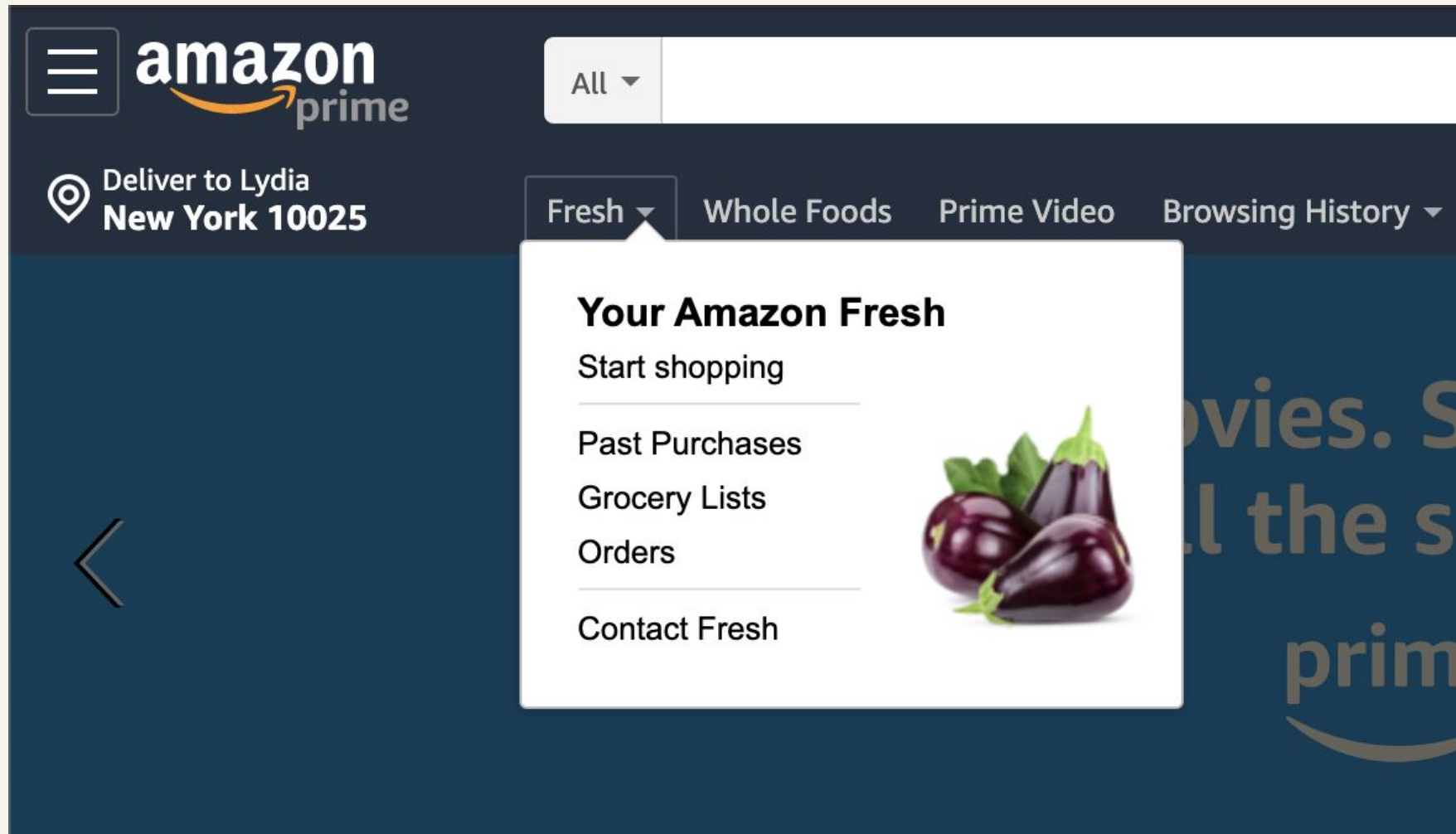
Time to move your
pointer to a target

$$= a + b * \log \left(\frac{2D}{S} \right)$$

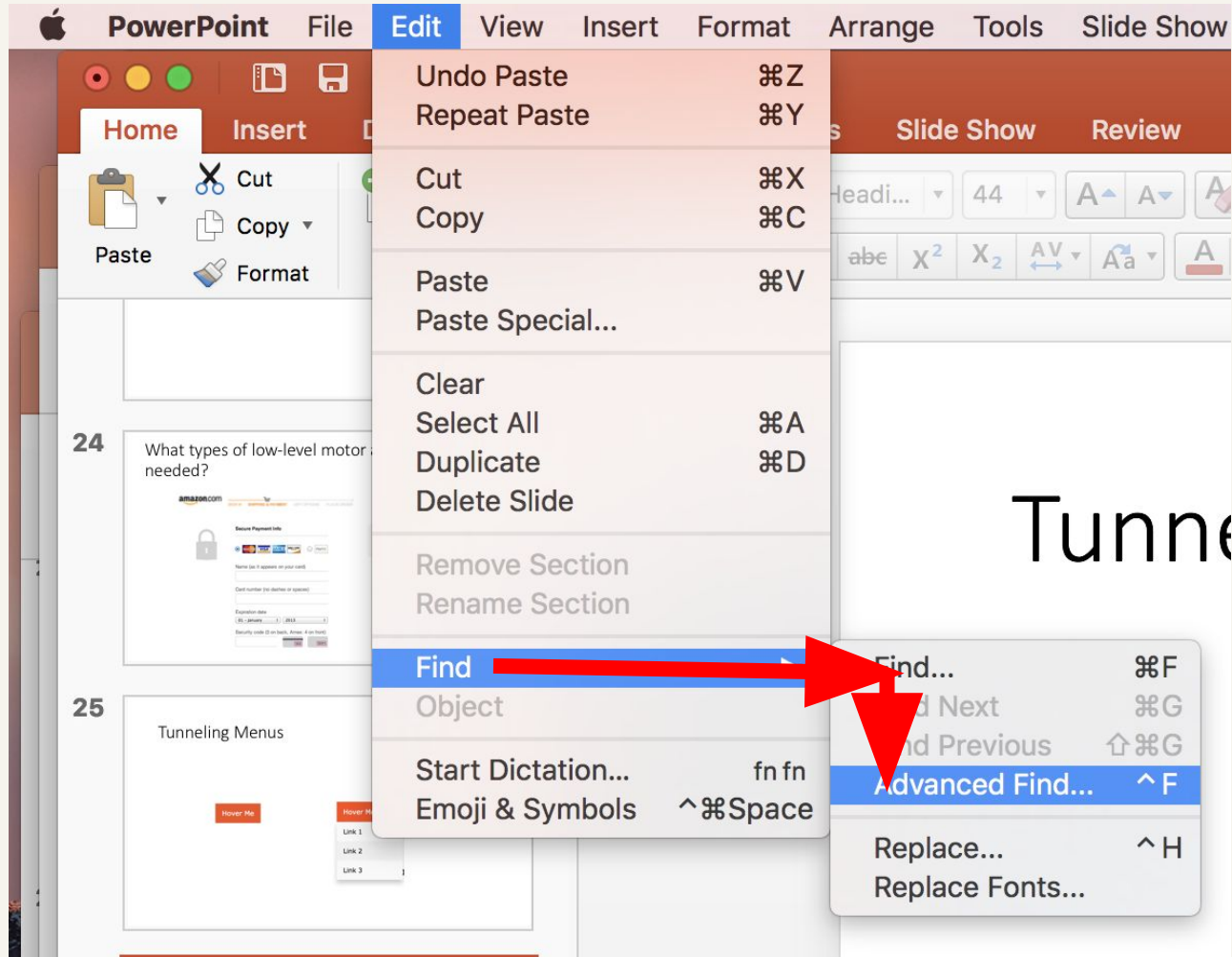


Buttons on the edges are
fast to get to because
they have an infinite size

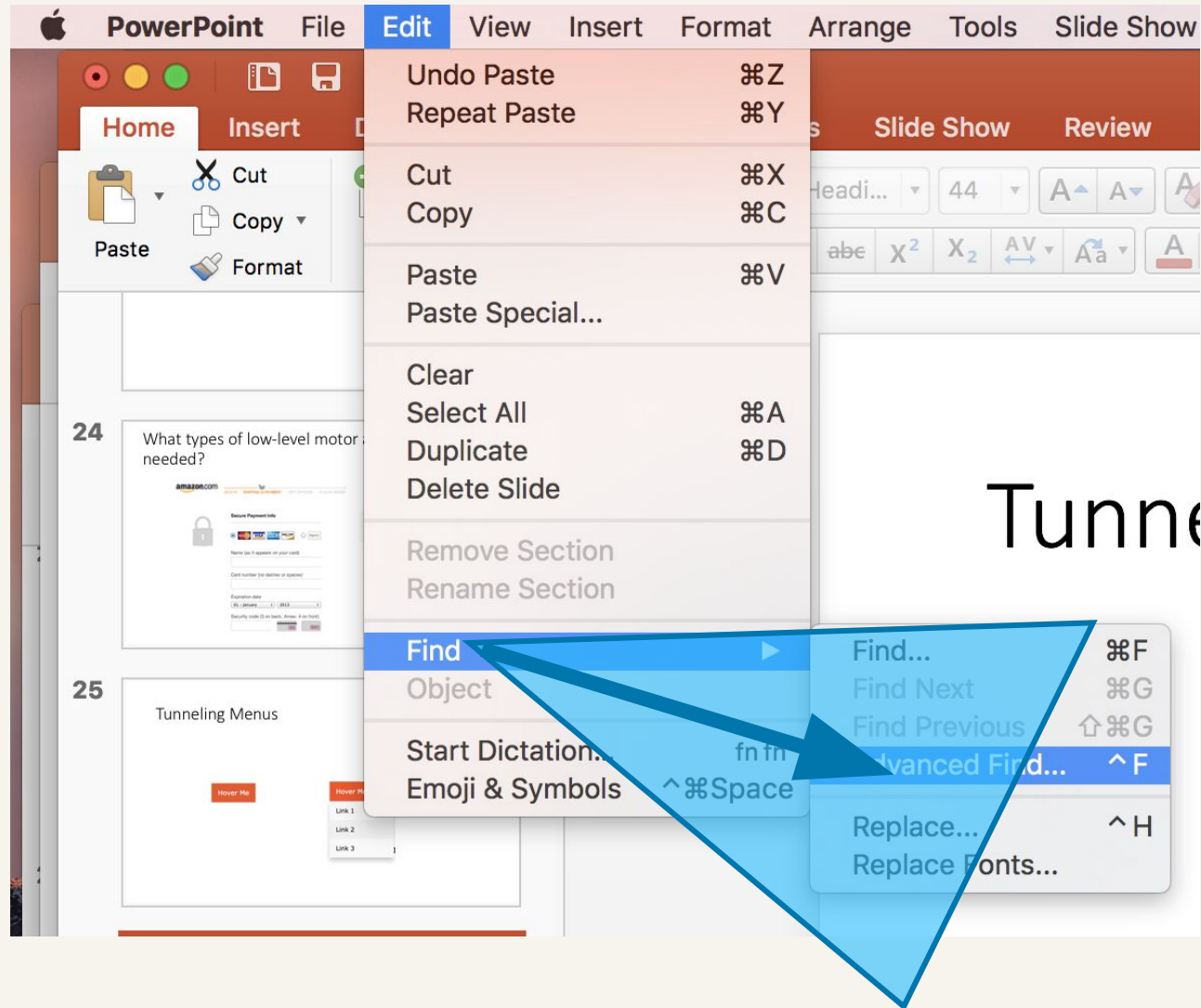
Moving & clicking: Tunneling Menus



Moving & clicking: Cascading Tunnel Menus



Moving & clicking: Cascading Tunnel Menus fix



What are all the low-level interactions needed to accomplish this subgoal?



Secure Payment Info



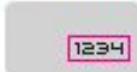
Name (as it appears on your card)

Card number (no dashes or spaces)

Expiration date

Security code (3 on back, Amex: 4 on front)





1. Click Card or Paypal
2. Click on name input field
3. Type name
4. Click on card number input field
5. Type card number
6. Click on expiration date menu
7. Select from date menu
8. Click on year menu
9. Select from another menu
10. Click on the security code input field
11. Type the security code
12. Press OK



Secure Payment Info



Name (as it appears on your card)

Card number (no dashes or spaces)

Expiration date

01 January 2013

Security Code (3 on Amex, 4 on others)

Every interaction takes time and effort,
and is a potential source of error.

Move

Click

Move

Click

TypeTypeTypeType

Move

Click

TypeTypeTypeType

Tunnel

Move

Tunnel

Click

TypeTypeTypeType
e

How could you improve this?

amazon.com

SIGN IN  SHIPPING & PAYMENT GIFT OPTIONS PLACE ORDER



Secu



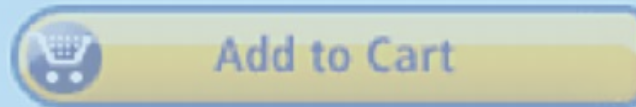
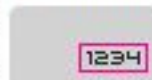
Na

Card

Exp

01

Security code (3 on back, Amex: 4 on front)



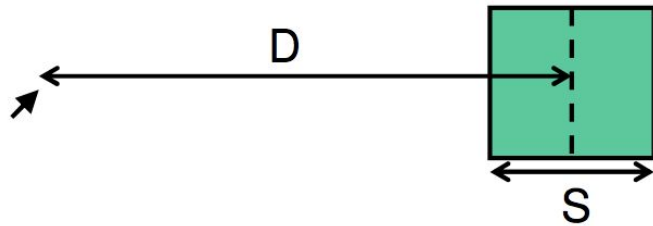
or 1-Click Checkout



Continue

review this order before it's final.

Low-level Interactions take time and effort. Minimize them because you do them a lot.



Secure Payment Info



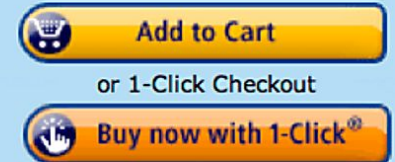
Name (as it appears on your card)

Card number (no dashes or spaces)

Expiration date

01 - January 2013

Security code (3 on back, Amex: 4 on front)



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The Interaction Loop



Establish a goal: Purchase a book.

amazon

Try Prime

All ▾ design of everyday things

Q

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EN

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Refine by

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☐ fresh

Delivery Day

☐ Get It Today

☐ Get It by Tomorrow

Amazon Prime

☐ prime

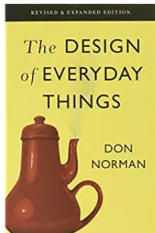
Eligible for Free Shipping

☐ Free Shipping by Amazon

Book Language

☐ English

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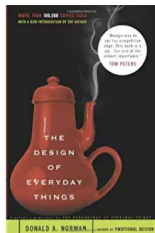
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Sep 19, 2002

by Donald A. Norman

Paperback

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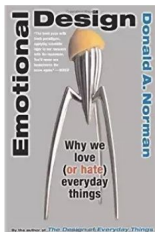
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Emotional Design: Why We Love (or Hate) Everyday Things

May 11, 2005

by Don Norman

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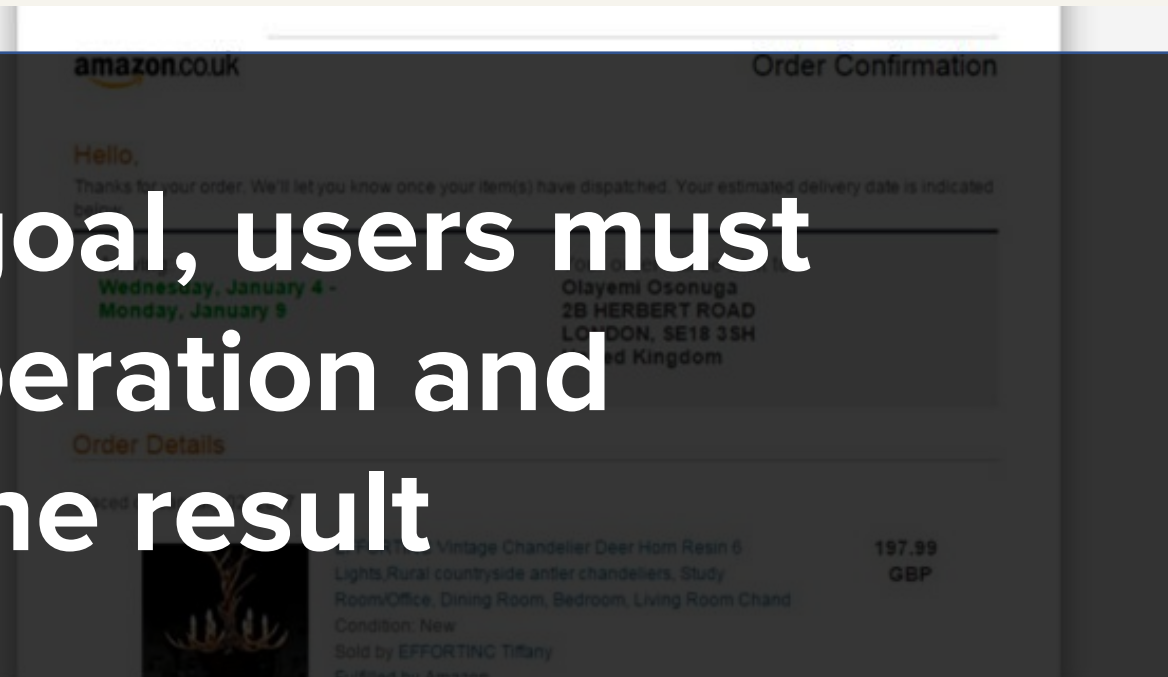
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★★★★☆ 61

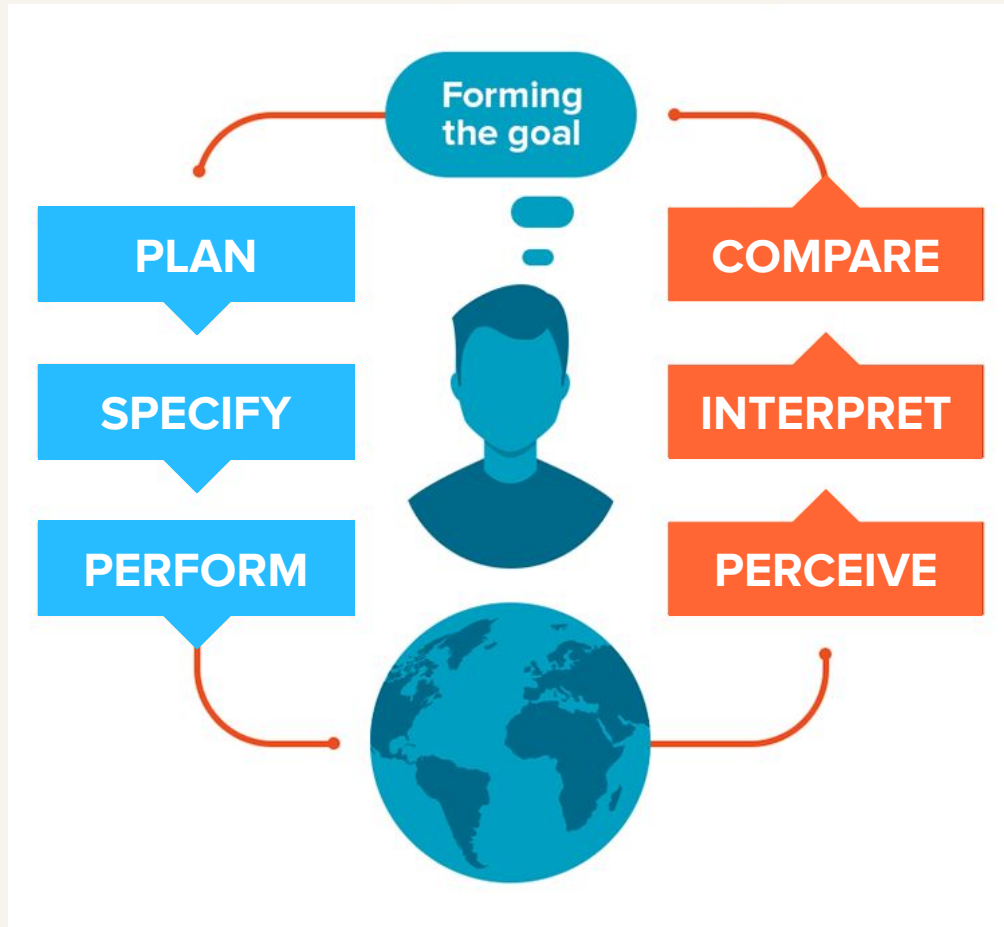


What happens after you complete your purchase?

**To accomplish a goal, users must
execute an operation and
evaluate the result**



The Seven Stages of Action



1. Form the goal

2. Plan the action

3. Specify the action sequence

4. Perform the action sequence

5. Perceive the state of the world

6. Interpret the perception

7. Compare the outcome with the goal



Establish a goal: Read at night.



Goal Execution Step 01:

amazon

Try Prime

Books ▾ design of everyday things

Q

Shop Valentine's Day Deals

Departments ▾

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EN

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DON NORMAN

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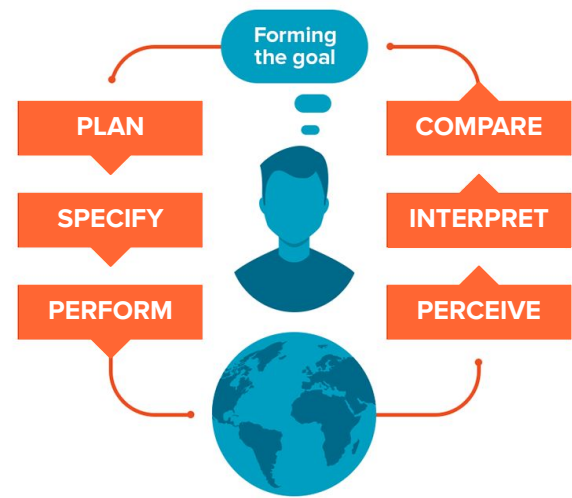
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Ship to: newyork, 10001 ▾

Add to List



Plan the action
Specify the action
sequence
Perform the action
sequence



Goal Execution Step 02:

amazon

Try Prime

Books ▾ design of everyday things

Q

Shop Valentine's Day Deals

Departments ▾

Your Amazon.com

Today's Deals

Gift Cards

Registry

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EN

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The DESIGN of EVERYDAY THINGS

DON NORMAN

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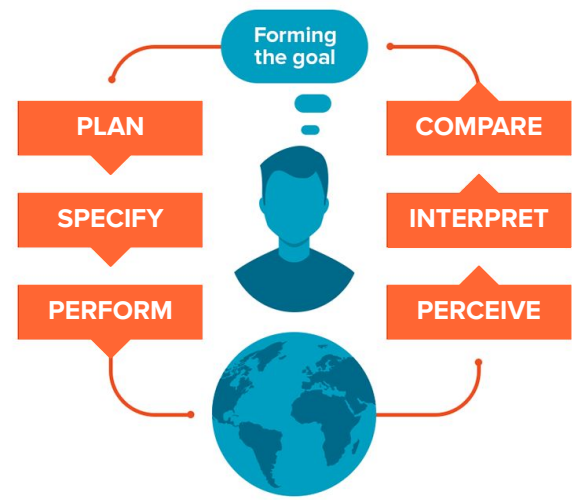
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Add to List



Plan the action

Specify the action sequence

Perform the action sequence



Goal Execution Step 03:

amazon

Try Prime

Books ▾ design of everyday things

Q

Shop Valentine's Day Deals

Departments ▾

Your Amazon.com

Today's Deals

Gift Cards

Registry

Sell

Help

EN

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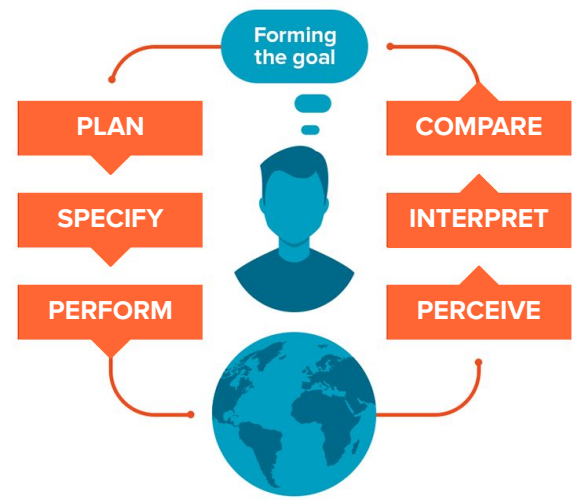
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Add to List



Plan the action
Specify the action
sequence
Perform the action
sequence



Goal Evaluation Step 01:

amazon.co.uk

Order Confirmation

Hello,

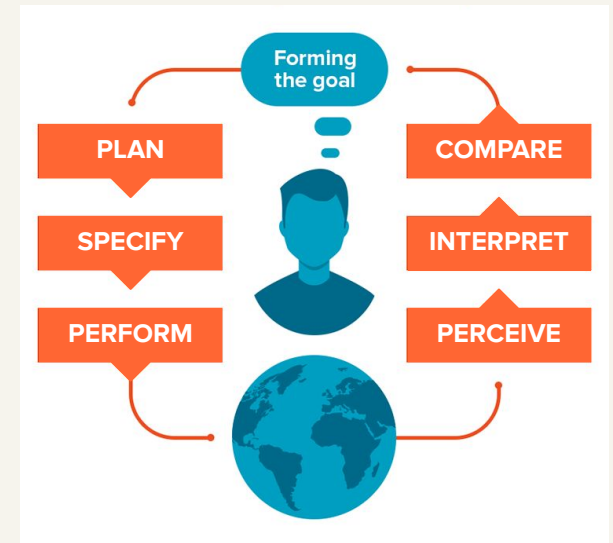
Thanks for your order. We'll let you know once your item(s) have dispatched. Your estimated delivery date is indicated below.

Arriving: Wednesday, January 4 - Monday, January 9	Your order will be sent to: Olayemi Osonuga 28 HERBERT ROAD LONDON, SE18 3SH United Kingdom
--	---

Order Details

Placed on January 02, 2017

	EFFORTINC Vintage Chandelier Deer Horn Resin 6 Lights,Rural countryside antler chandeliers, Study Room/Office, Dining Room, Bedroom, Living Room Chand Condition: New Sold by EFFORTINC Tiffany Enlisted to Amazon	197.99 GBP
---	--	-----------------------



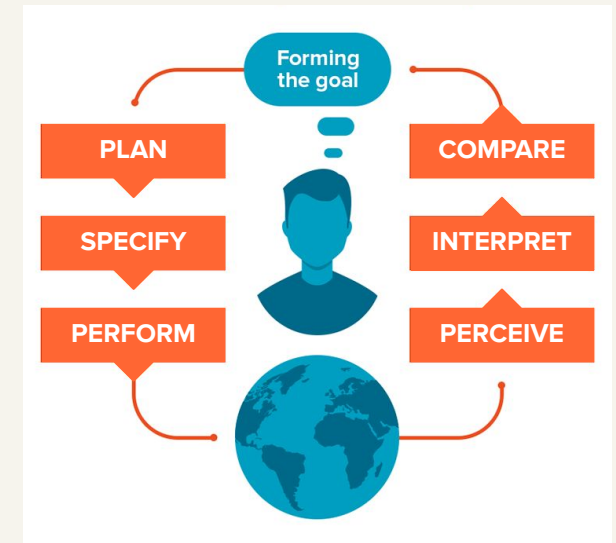
**Perceive the State
of the world**

Interpret the
perception

Compare the
outcome with the
goal



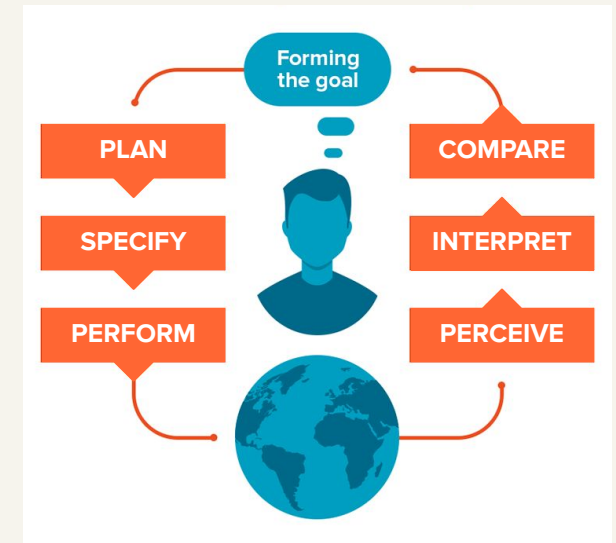
Goal Evaluation Step 02:



Perceive the State
of the world
**Interpret the
perception**
Compare the
outcome with the
goal



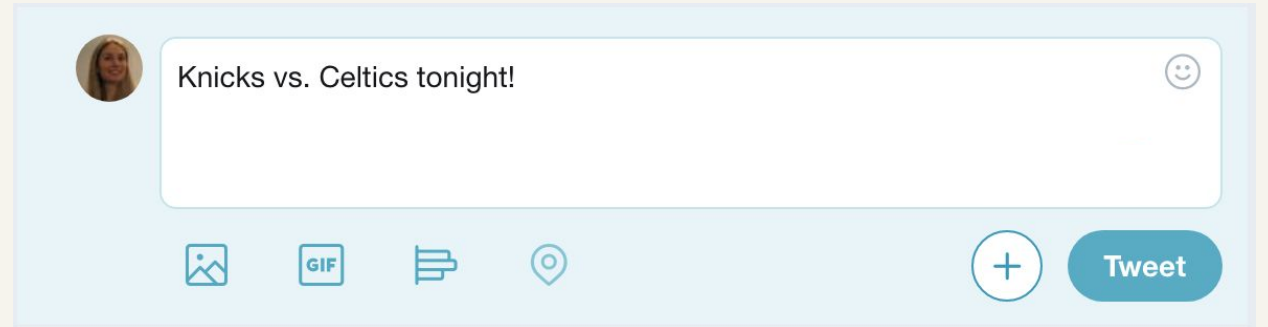
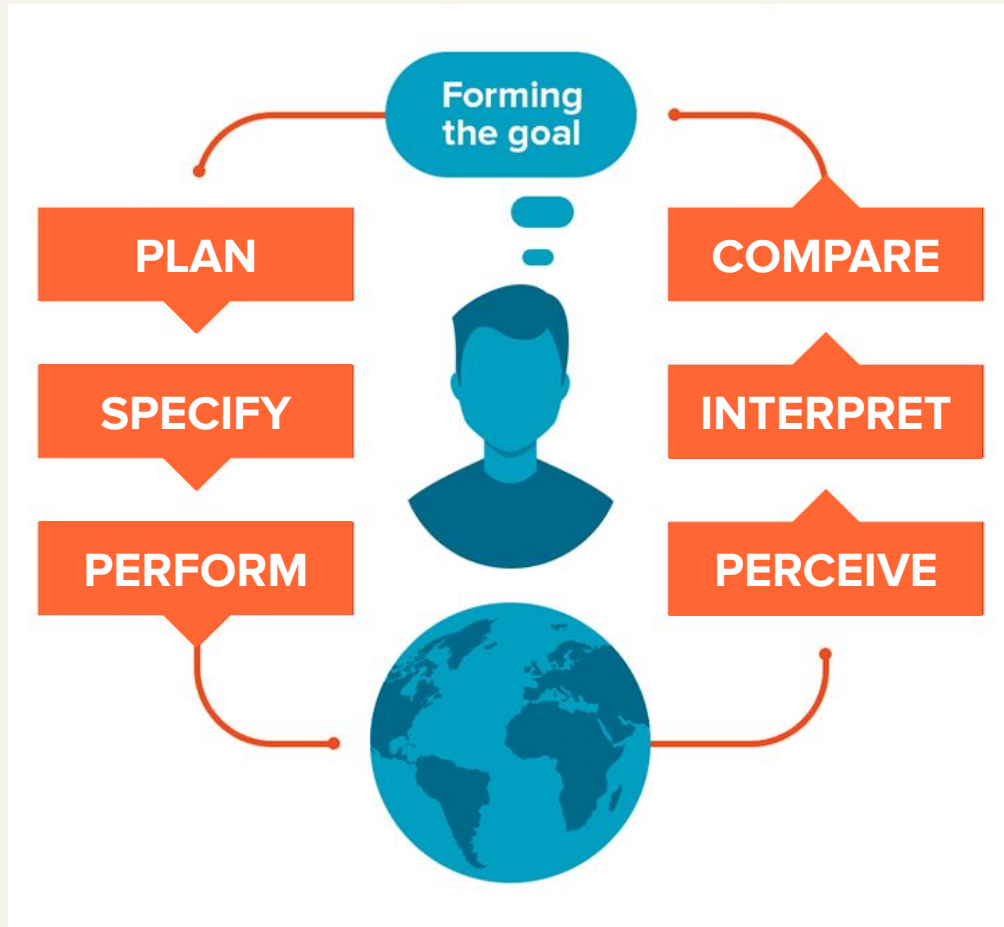
Goal Evaluation Step 03:



Perceive the State
of the world
Interpret the
perception
**Compare the
outcome with the
goal**



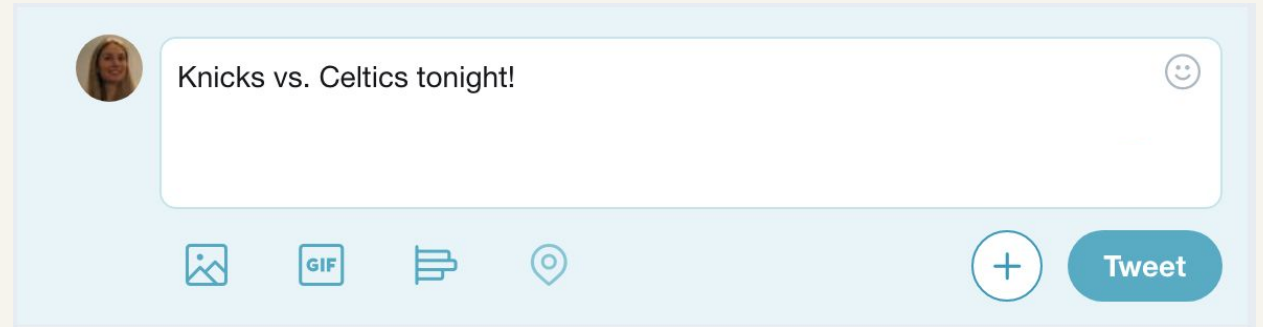
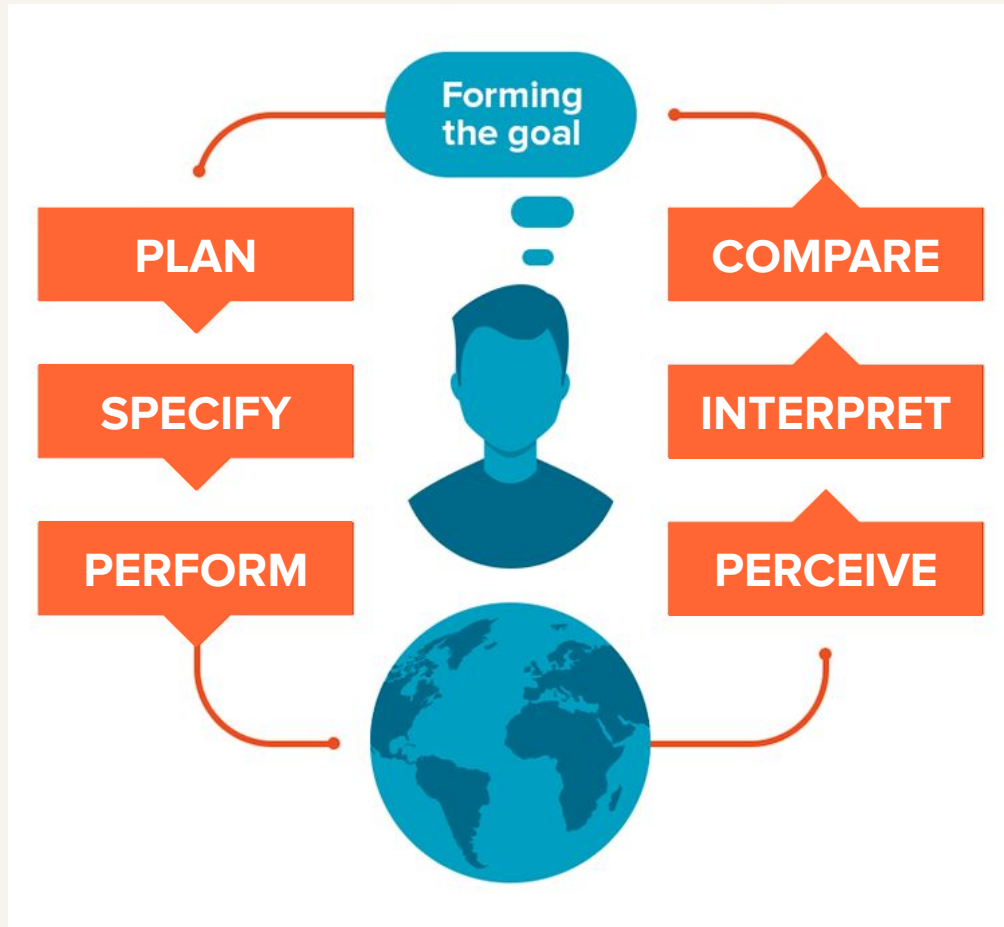
What's the users goal? Post a tweet



Post a tweet



What does this execute?



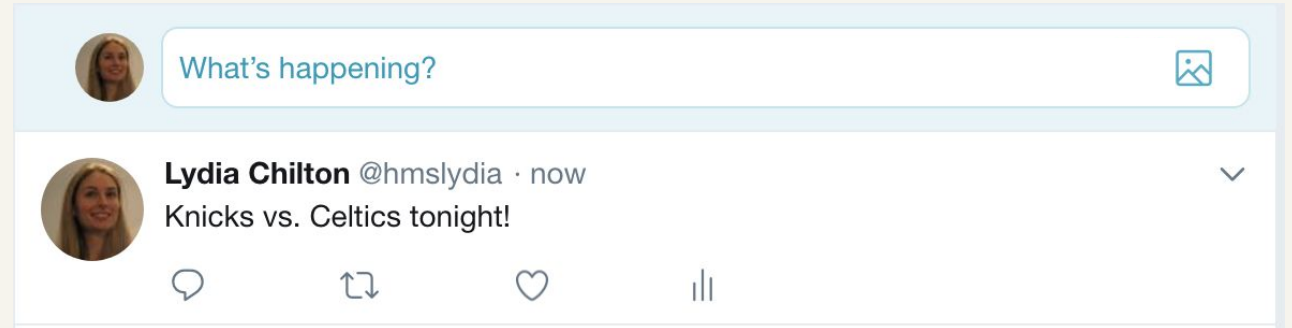
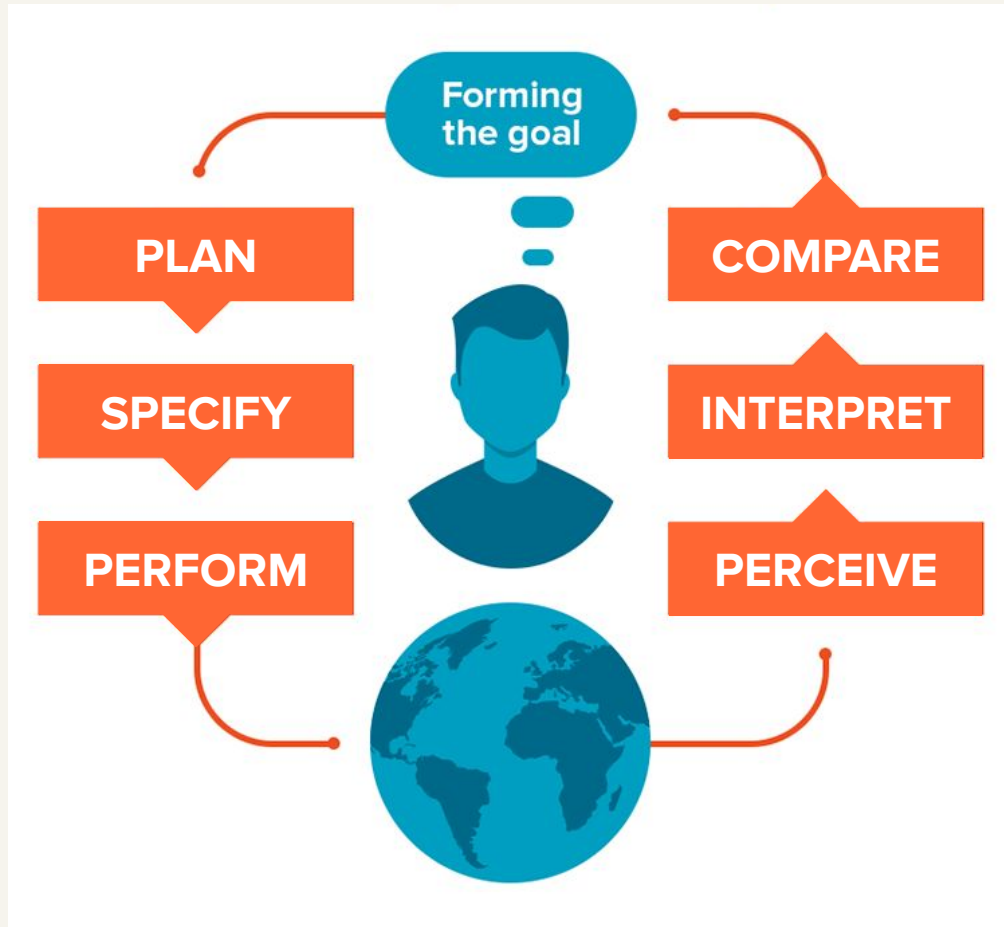
Put cursor in box

Type message

Move mouse to button and click



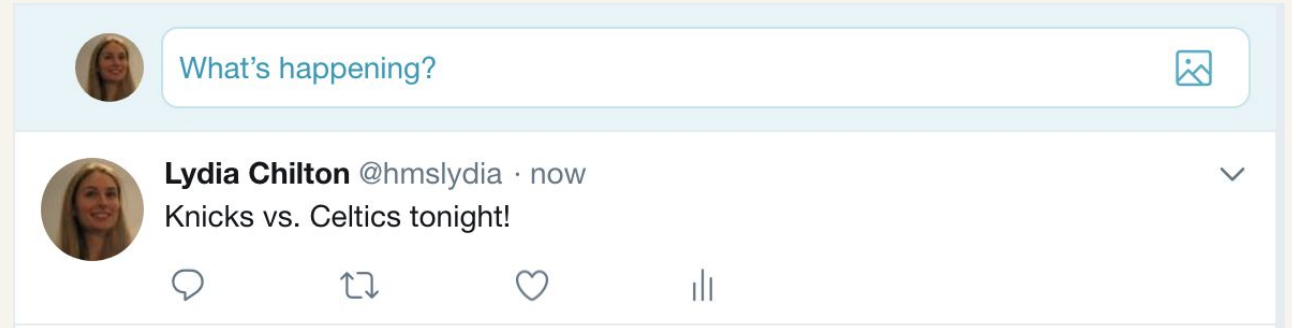
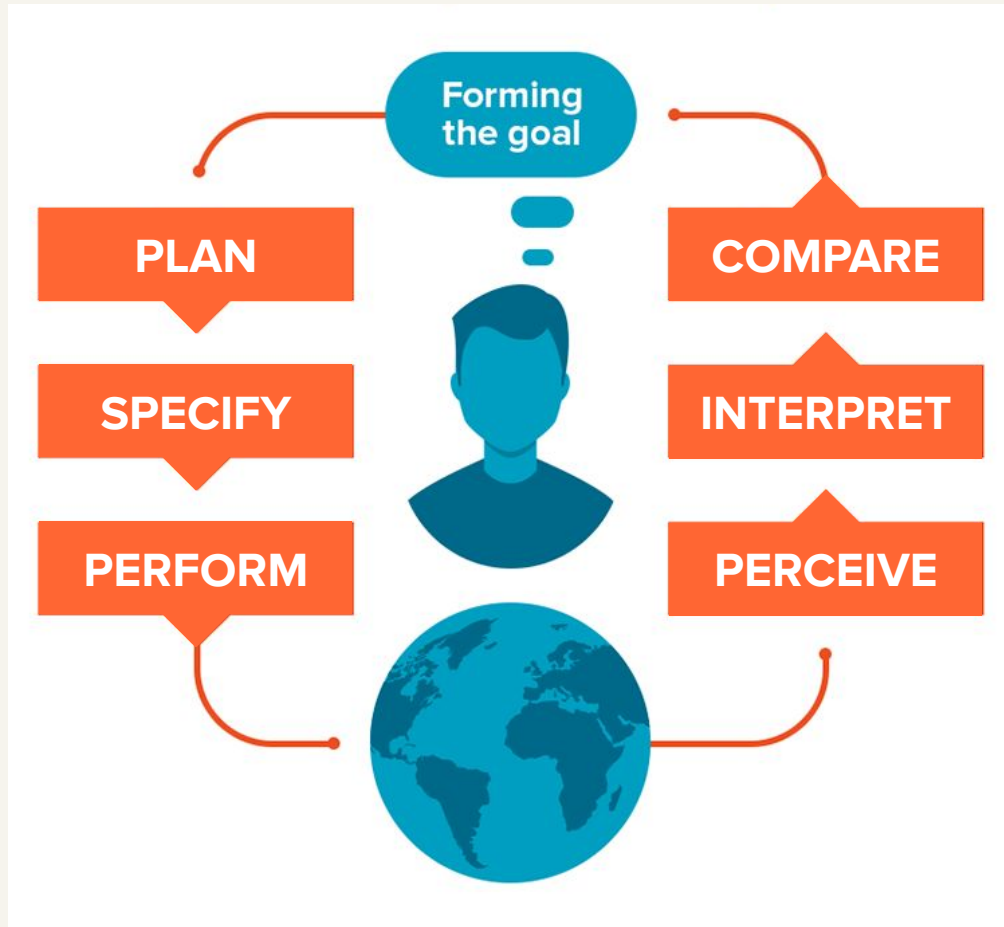
What does this execute?



Did it get posted?



How does the user know?



It's Lydia Chilton's face

It's text that she wrote

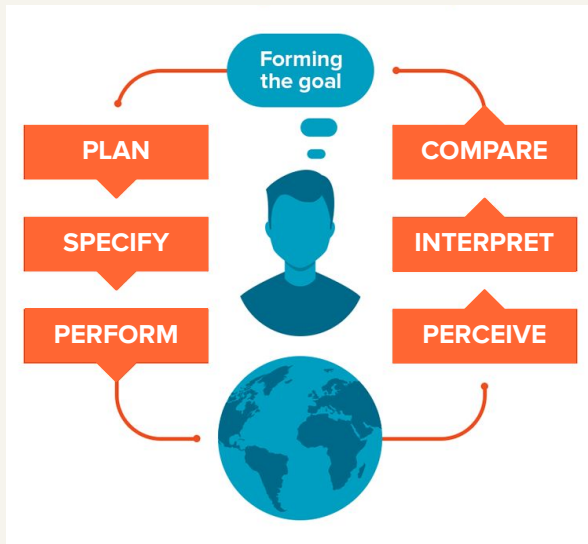
It has new options



Goal: Perfect Dance Dance Revolution Score

What is the users' subgoal?

Step on the correct arrow at the correct time.

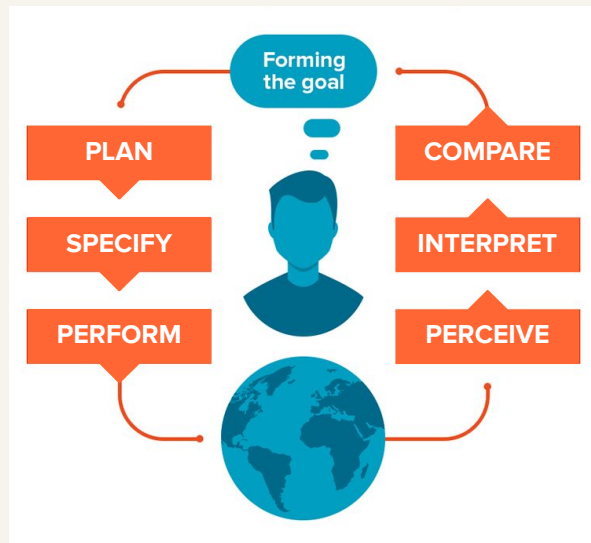


Forming the goal



Goal Execution Step 01: How does the user plan the action?

Look at the screen to see the
correct arrow / timing

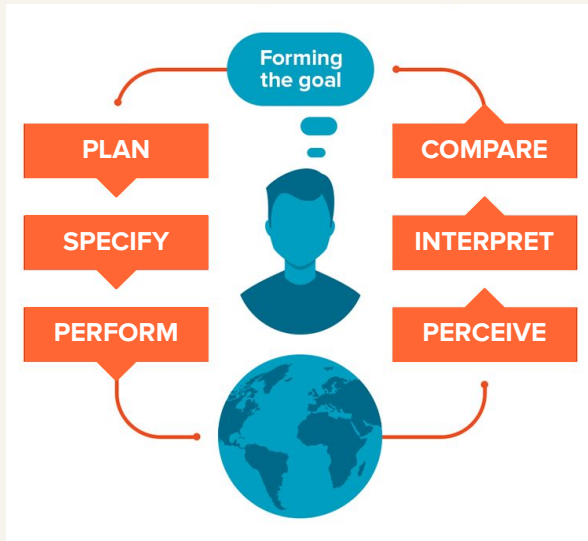


Plan the action
Specify the action
sequence
Perform the action
sequence



Goal Execution Step 02 & 03: How does the user execute the action?

Lift your foot, move over the
arrow, step on it at the right
time.

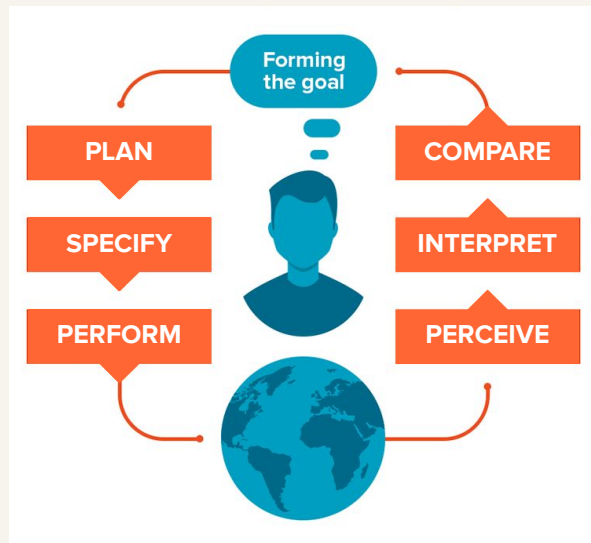


Plan the action
**Specify the action
sequence**
**Perform the action
sequence**

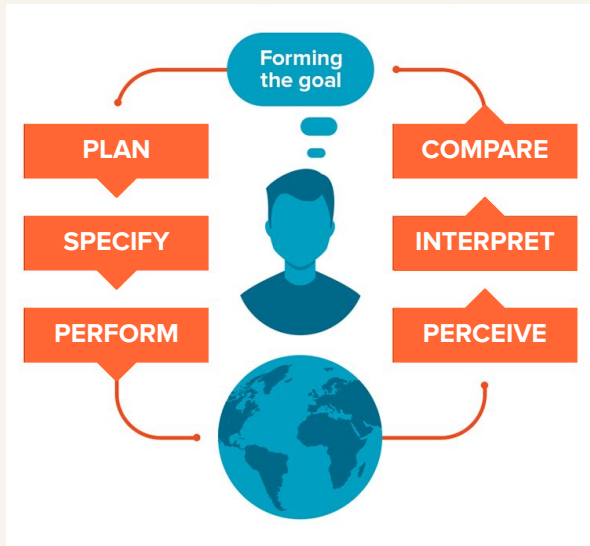


Evaluation: How does the user evaluate the action?

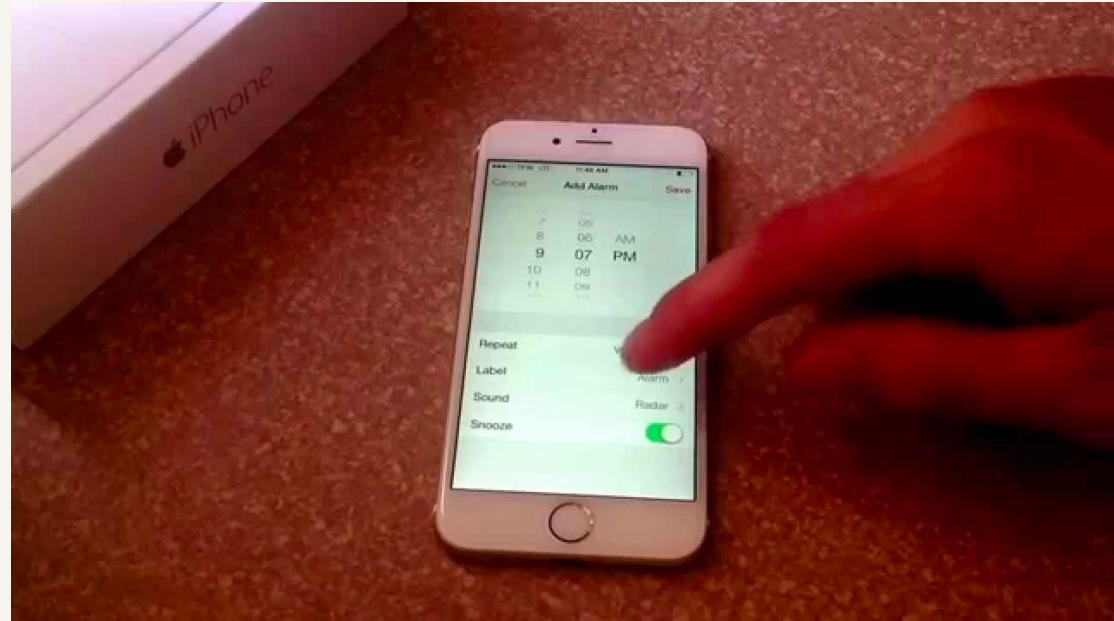
You can see the arrow flash
It tells you a grade



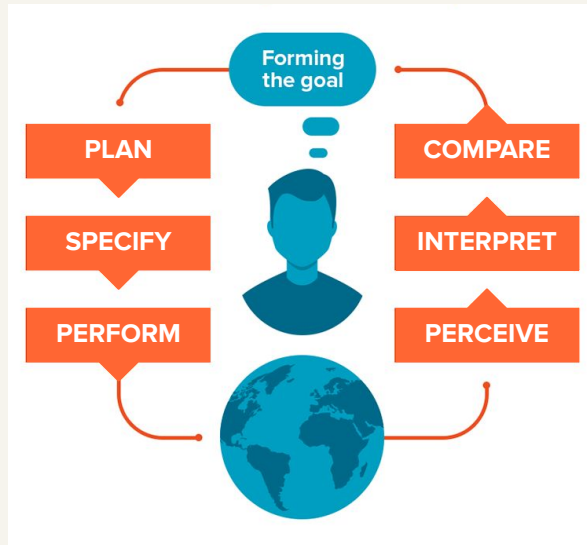
What is the user's goal? Set the phone alarm for 9:07PM



To set the alarm for 9:07PM

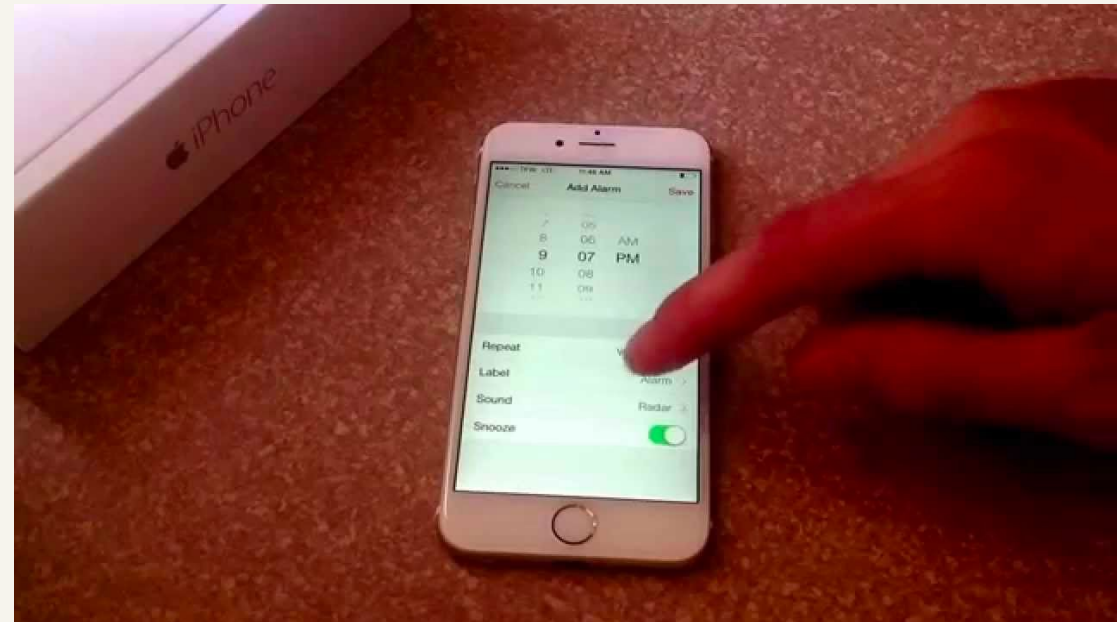


Goal Execution?

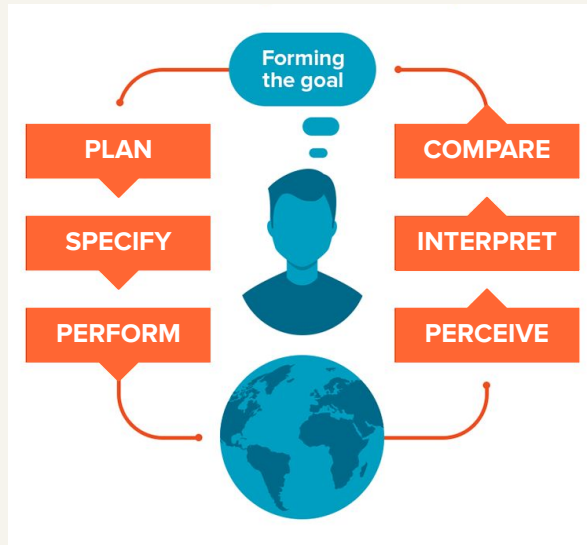


Move the wheel to the chosen time

Switch the alarm to “on”



Goal Evaluation?



Turns to an alarm screen

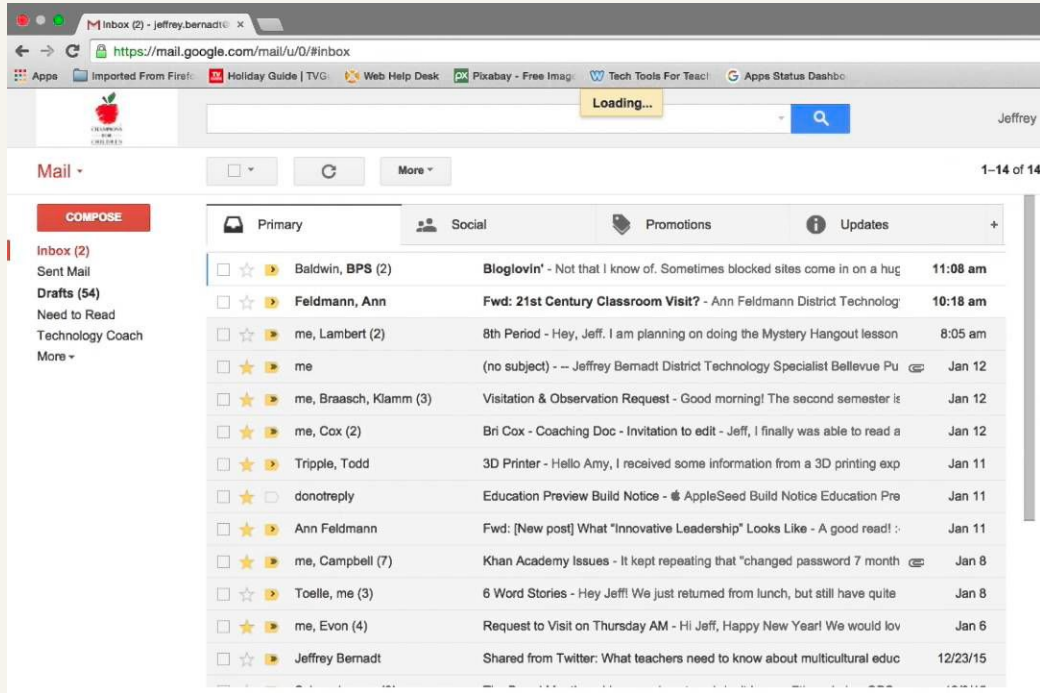


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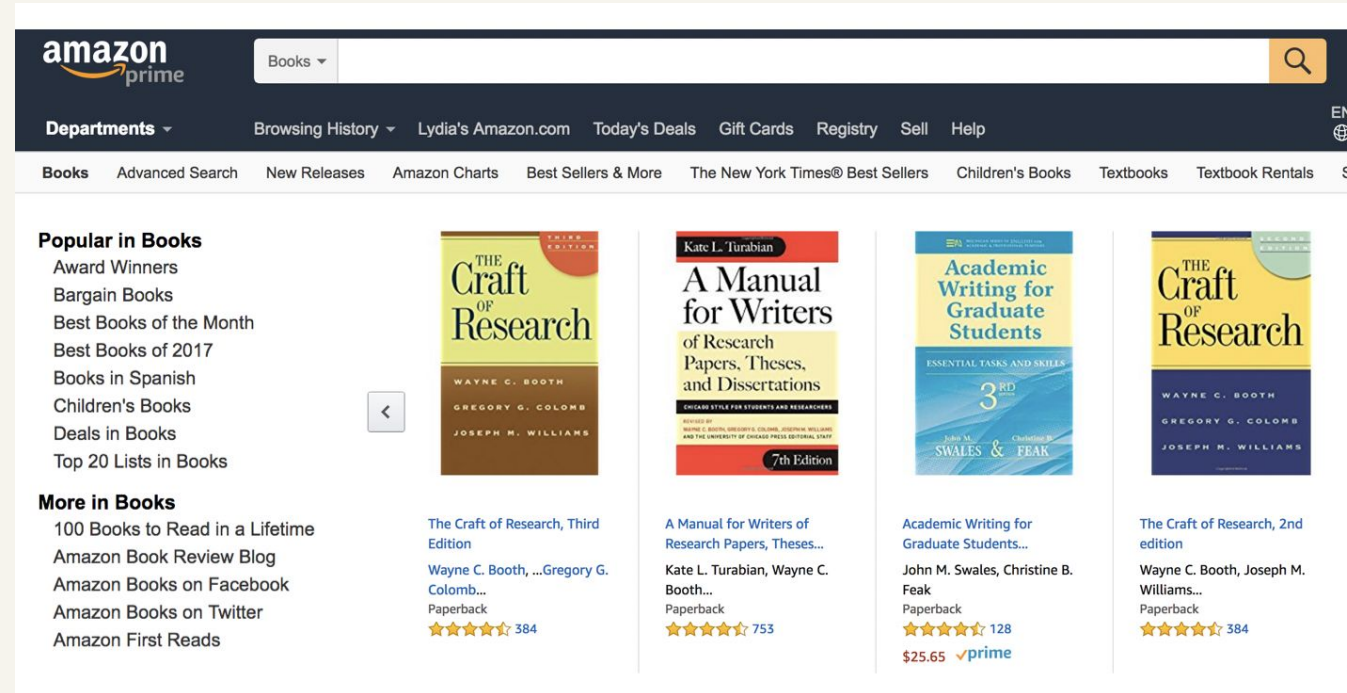
Summary



Users interact with a system to accomplish a goal



To read and respond to all email.



To purchase a book.



The designer must create the subgoals and interactions to help users accomplish it.

Goal: Purchase a book

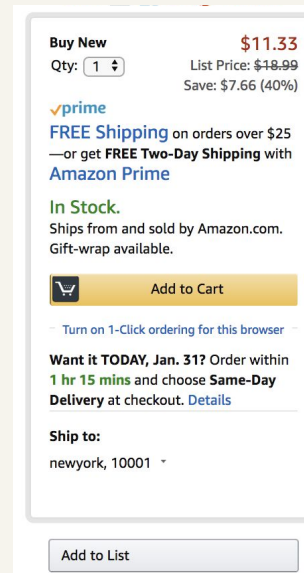
Subgoal: Find it

Interaction: *Type and click*



Subgoal: Add to cart

Click



Subgoal: Enter payment info

Type, click, point

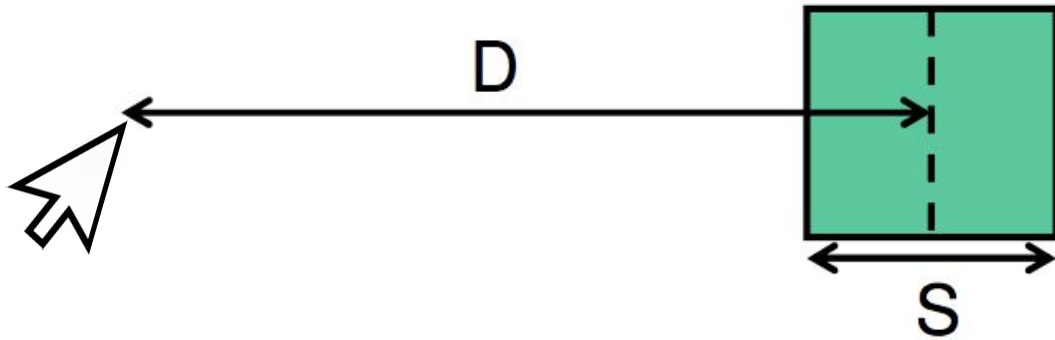
A screenshot of the Amazon checkout page. It shows fields for "Name (as it appears on your card)", "Card number (no dashes or spaces)", and "Expiration date". There are also fields for "Security code (3 on back, Amex: 4 on front)". The page includes logos for various payment methods: MasterCard, VISA, AMEX, DISCOVER, and PayPal.

Subgoal: Place order

Click

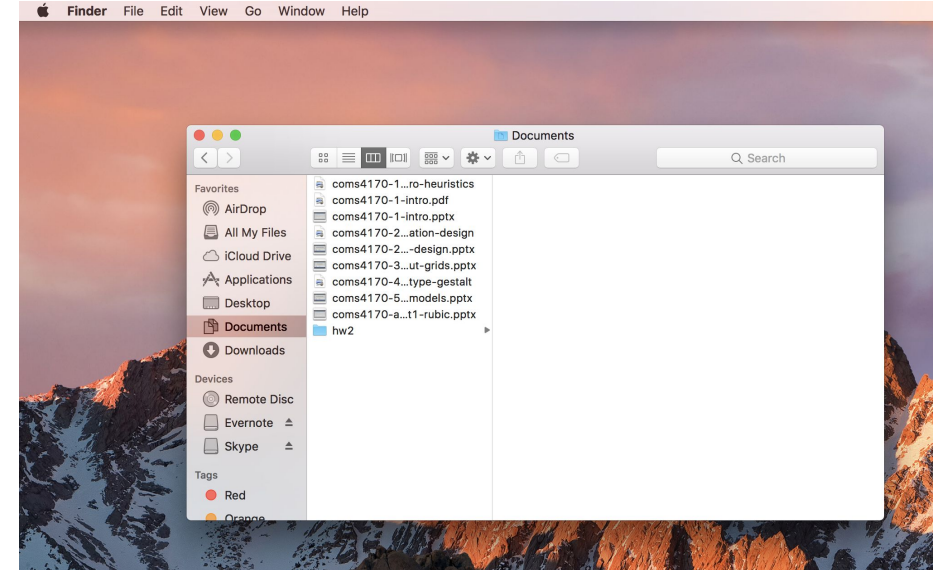
A screenshot of the Amazon "Place your order" page. It features a "Place your order" button at the top. Below it is an "Order Summary" section showing the items, shipping and handling costs, and the total order amount of \$25.37. The summary also includes tax and rewards points information. At the bottom, there is a section titled "How are shipping costs calculated?" explaining that Amazon Prime Shipping has been applied to eligible items.

Fitts's Law: Time to move the pointer



Time to move your
pointer to a target

$$= a + b * \log \left(\frac{2D}{S} \right)$$



Buttons on the edges are
fast to get to because
they have an infinite size

**Low-level interactions take time and effort.
Minimize them because you do them a lot.**



Secure Payment Info



Name (as it appears on your card)

Card number (no dashes or spaces)

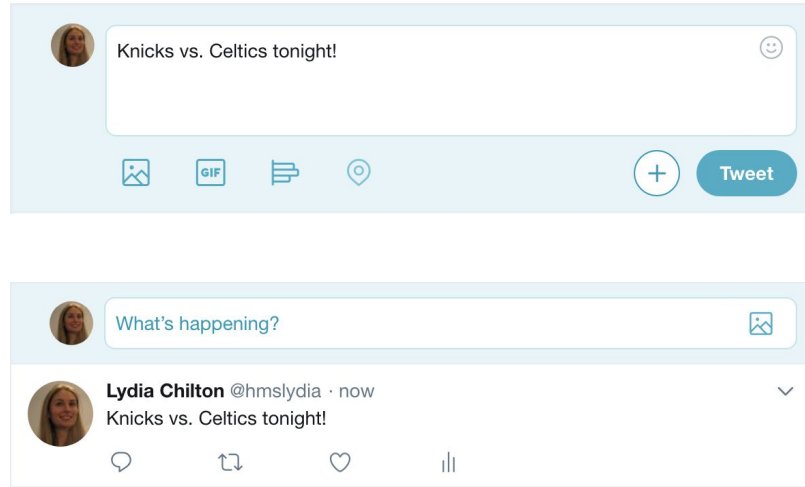
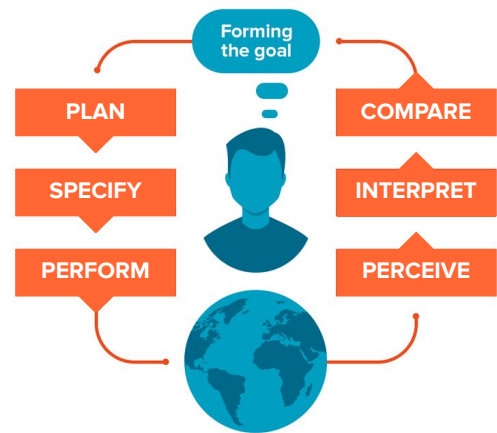
Expiration date

01 - January 2013

Security code (3 on back, Amex: 4 on front)

Move
Click
Move Click
TypeTypeTypeType
Move Click
TypeTypeTypeType
Move Tunnel Click
Move Tunnel Click
TypeTypeTypeType

Know the users' goals and design interactions as: execution and evaluation



1. Form the goal

Execution

2. Plan the action
3. Specify the action sequence
4. Perform the action sequence

Evaluation

5. Perceive the state of the world
6. Interpret the perception
7. Compare the outcome with the goal

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Homework 02



Homework 02

Warm Up due **Friday at 11:59PM**

- You can complete this now

Main Assignment due **Monday at 11:59PM**

- You will need Tuesday's lecture

Homework is posted to Courseworks:

<https://courseworks2.columbia.edu/courses/237745>



